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Surgical Simulation: Virtual Reality to Artificial Intelligence



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Introduction

“See one, do one, teach one” is an adage no longer applicable to most modern academic medical centers which have embraced instead the idea of “perfect practice makes perfect.” It is no longer acceptable to use patients to practice on to learn to throw your first endostitch or create your first stapled anastomosis. These skills should be practiced and mastered in a basic skills lab or a comprehensive simulation center and then brought to the bedside to provide quality care as part of a team.

Simulation was once novel and is now commonplace. After a simulation lecture to Boston Surgical in 2005, the chairmen of surgery at Brigham and Women's Hospital (BWH) and Massachusetts General Hospital (MGH) challenged whether we needed to change an educational process that was not broken. Drs. Zinner and Warshaw were correct that simulation was expensive, required space, and took trainees out of the operating theaters.

By 2009 when *Current Problems in Surgery* published “Surgical Skill Training and Simulation,”¹ all Boston teaching hospitals were fully engaged in simulation training, innovation, and education research. That issue outlined the transition from learning on patients and using vivarium and cadaver labs to using video trainers and simulation. We showed how training on simulators improved operative performance of surgical residents performing a laparoscopic hernia repair (Fig 1)² and LC (Fig 2).³ In this issue of *Current Problems in Surgery*, we highlight advances in simulation over the last 25 years.

Since 2009, the American College of Surgeons (ACS) Accredited Education Institutes (AEI) have grown from 27 to more than a hundred accredited centers worldwide as more medical

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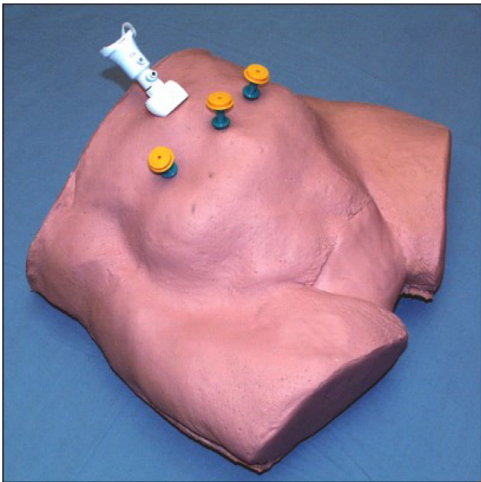
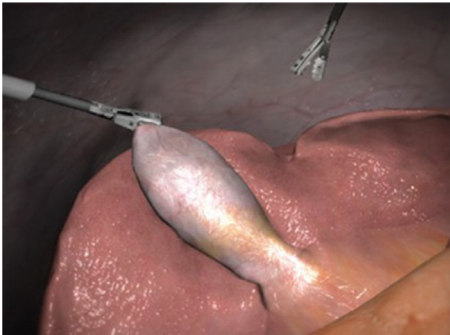
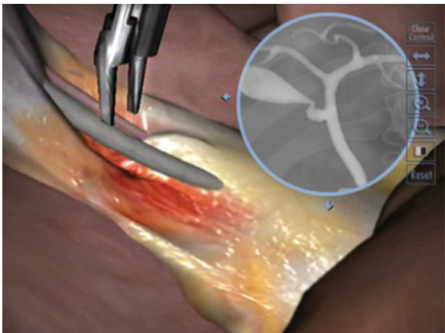


Fig. 1. Laparoscopic hernia model.

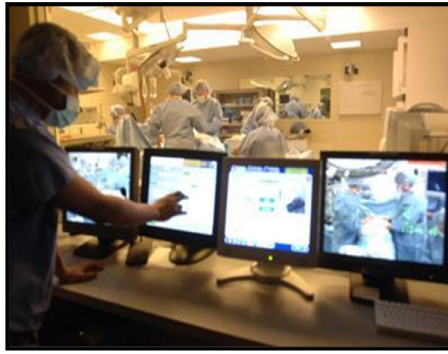


A



B

Fig. 2. Virtual reality (VR) laparoscopic cholecystectomy: (A) retraction of gallbladder, (B) clip cystic duct.



A



B

Fig. 3. Team training using simulation scenarios. (A) Control room. (B) Debrief.

schools and training programs invest in simulation centers and adopt skills-based training and team training. The Fundamentals of Laparoscopic Surgery (FLS) and Fundamentals of Endoscopic Surgery (FES) are required by the American Board of Surgery (ABS) before taking the general surgery qualifying examination. Gynecology also requires FLS. The Association for Surgical Education (ASE) and Association of Program Directors in Surgery (APDS) have popularized their curricula for medical students and residents, respectively.

Simulation technology has quickly evolved from early box trainers to more and more realistic virtual reality (VR) simulators.^{4,5} This is particularly true for VR simulators to teach robotic skills and operations such as robotic ventral and inguinal hernia.

With experience, simulators achieved higher realism with scenarios. Team interactions and performance were being measured. Inspired by the work of Lord Ara Darzi and colleagues at Imperial College, London, Dan Jones built the first mock endosuite in 2006 at the Carl J. Shapiro Simulation and Skills Center at Beth Israel Deaconess Medical Center. The endosuite includes multiple camera mountings, directional microphones, and sophisticated data management to record all communication and activity during a laparoscopic simulation. A separate control room with video monitors overlooked the simulation via a 1-way mirror. After simulation scenarios, video clips are reviewed and discussed in the Debrief Room (Fig 3). Seasoned surgeons could be compared to younger surgeons.^{6,7} Atul Gawande successfully lobbied CRICO, the Harvard mal-practice carrier, to fund all surgeons to experience “OR Team Training with Simulation” across the Harvard teaching hospitals (Beth Israel Deaconess Medical Center [BIDMC], MGH and BWH,

and Children's Hospital). The curriculum emphasized closed loop communication, speaking up, and the use of checklists.⁸ Ten years later, malpractice cases related to poor communication have trended down. Nevertheless, many wondered whether team training might be better taught with VR.⁹

The challenge for engineers was whether they could create a virtual operating room (OR) as a safe place to make errors and learn as individuals and as teams. Suvranu De at Rensselaer Polytechnic Institute and Dan Jones at BIDMC created a VR OR.¹⁰ The surgeons wore an Oculus headset and could interact through an avatar with the virtual OR. The first study measured the response of surgeons, anesthesiologists, and nurses to an OR fire.¹¹ Less than 5% of the OR team responded appropriately and turned off the oxygen from the ventilator. After repetition in the VR OR, most learned the correct steps. For those who struggled, the simulator guided the avatar to make the correct, safe choice, thereby using artificial intelligence (AI) guidance to help in decision making in crisis.

Most recently, we have used the virtual OR to better understand surgeon behavior. At the Society of American Gastrointestinal and Endoscopic Surgeons (SAGES) annual conference in 2024, participants wore an Oculus Headset with audio. The participant was engaged with 3 other confederate avatars during an acute hemorrhage scenario during laparoscopy. During the crisis, the other 3 avatars introduced verbal microaggressions during the operative procedure. Afterwards, the participant surgeons were asked about what they heard and did, or more commonly did not do, to address microaggressions. We were now able to use the safe space of VR OR to teach professionalism.

As we use multi-avatar simulation platforms, it is likely that AI will learn how we think as individuals and as teams. AI will not only help us get to the right answer, AI will help us do the right thing. Embracing technology has the potential of improving patient safety. Herein, we again restate the importance of training surgeons using simulation for both technical and nontechnical skills. We update the exciting advances in simulation over the last 16 years since our last publication in *Current Problems in Surgery*, such as VR¹⁰, AI¹¹ and 3-D printing.¹² We reimagine the future with the virtual mock OR and role of AI in decision making in crisis scenarios.

Credentialing of programs and surgeons

The ACS established the accreditation of simulation centers in 2005. The key elements required are space, faculty, personnel, budget, innovation, and the use of adult education theory and feedback. By 2009, 27 simulation centers were accredited. Today nearly 100 centers are accredited comprehensive simulation centers. However, many others are significantly under-resourced by medical schools and hospitals.

The ability of simulation to enhance surgical education and improve patient outcomes has been repeatedly demonstrated over time. As a result, simulation is now formally required for accreditation of surgical training programs and credentialing of surgeons. The Accreditation Council for Graduate Medical Education (ACGME) explicitly requires simulation facilities, experiences, and/or administrative resources for many procedural specialties. For general surgery, the ACGME Program Requirements¹³ state that "programs must provide for simulation and skills laboratories." Furthermore, they require personnel for administration of simulation, "a level-specific, simulation-based curriculum that complements clinical rotations in the development of technical and nontechnical skills," and a designated "individual to manage the portfolio of simulation activities." For obstetrics and gynecology, the ACGME requires "space and equipment for the educational program including...access to simulation resources" and includes "simulation training" among the required "structured and regularly held" educational sessions.¹⁴ The ACGME requires simulation facilities for other procedural specialties, with the orthopedic surgery requirements including "a dedicated space to facilitate basic surgical skills training,"¹⁵ and plastic surgery requirements stating that plastic surgery "programs must provide for skills laboratories."¹⁶ The ACGME also requires that residents in an anesthesiology training program "must participate in

at least one simulated clinical experience each year.”¹⁷ These explicit program requirements ensure simulation is universally available to trainees in these specialties.

Simulation has also been formally included in the certification process for individual surgeons. Beginning in 2010, the ABS has required successful completion of several programs that utilize simulation-based training and assessment, namely Fundamentals of Laparoscopic Surgery (FLS), Advanced Cardiac Life Support (ACLS), and Advanced Trauma Life Support (ATLS), for applicants to attain eligibility for the general surgery qualifying examination.¹⁸ In 2014, the ABS introduced their Flexible Endoscopy Curriculum for General Surgery Residents, which culminates with certification in the simulation-based Fundamentals of Endoscopic Surgery (FES). Starting from the 2017-2018 academic year, FES certification was added to the certifications listed above as requirements for eligibility for the qualifying examination. The American Board of Obstetrics and Gynecology (ABOG) has required certification in FLS for its board certification for residents graduating after May 31, 2020. As of now, the ABS does not require any simulation-based assessment for its Continuous Certification process. However, there may be a role in the future for simulation-based assessment for maintenance of certification, although it would require development of standardized, validated assessment instruments. Beyond board certification, credentialing of individual surgeons to perform procedures usually occurs at the facility level.

Credentialing requirements often include board certification, which incorporates the simulation-based assessments for surgeons who completed training after their respective boards added those requirements. Because these simulation-based assessments may be the only objective measures of a surgeon's knowledge and skill within the respective practice domain, some have advocated for their use in credentialing for all surgeons who perform relevant procedures. In 2012, the ACS and SAGES issued a joint statement recommending FLS certification as an institutional credentialing requirement for all surgeons who perform laparoscopic surgery.¹⁹

Impact of the COVID-19 pandemic

Surgical simulation and the COVID-19 pandemic

The coronavirus disease 2019 (COVID-19) pandemic impacted essentially every aspect of medical care and training around the world. Like many other domains, surgical simulation experienced an initial cessation of activities as resources were diverted to acute clinical care and systems were put in place to assure participants' safety. However, although many routine simulation activities were canceled or postponed, simulation featured prominently within the pandemic's rapidly evolving clinical environment to hone newly developed protocols and procedures. Ultimately, through resumption of traditional simulation activities and development of new, often virtual, opportunities, simulation played a major role in the maintenance of skill throughout the pandemic period and overcoming the gap in clinical volume for residents and students whose training was interrupted by the pandemic.

At the onset of the pandemic in early 2020, many groups paused simulation and assessment activities in the face of social distancing requirements, personnel scheduling adjustments, and various unknowns related to disease transmission and health care system needs. SAGES closed FLS, FES, and Fundamental Use of Surgical Energy (FUSE) test centers from March 27, 2020, through May 31, 2020. On reopening of test centers in June 2020, they noted that local restrictions might still impact test center availability and offered remote testing options for institutions to administer the examinations at their own facilities. Recognizing the potential difficulty of obtaining those certifications, the ABS relaxed the requirements for their qualifying and certifying examinations in 2020 and 2021, allowing residents completing training in those years to take 1 or both exams without first receiving FLS and/or FES certification. However, applicants were still required to complete those certifications to achieve final board certification by the ABS. The ACS Committee on Trauma (CoT), recognizing the challenges of scheduling ATLS renewal courses amid the pandemic, offered extensions on ATLS certifications. Initially, they extended the certifi-

cation through 2020 for providers whose certification expired in late 2019 and who remained in the grace period and for 1 additional year for those expiring in 2020. Ultimately, they granted certification extensions into 2023 for ATLS students who took a mandatory content review and for ATLS faculty.

The ACS Division of Education appointed a special committee through the Academy of Master Surgeon Educators in March of 2020 to evaluate the impact of and response to the pandemic for surgical training. The committee conducted a survey of surgery chairs, program directors, and Academy members from April to June of 2020. The results showed that surgical simulation was significantly impacted early in the pandemic, with 44% of respondents reporting that simulation centers were not accessible and 53% of respondents reporting suspension of simulation training. At the same time, simulation emerged as a key theme among innovations implemented in response to the pandemic.²⁰ The special committee also conducted surveys of surgery clerkship directors in the spring of 2020 and 2021. Regarding access to simulation training, 42% of clerkship directors who responded to that item on the 2021 survey noted a negative impact of the pandemic whereas 49% reported no impact. However, virtual adaptation of simulation and skills training was the most-cited educational adaptation by clerkship directors.²¹

Adaptation of simulation programs during the pandemic

As the pandemic progressed, surgical educators recognized the need to adapt traditional simulation methods to provide surgical skills training across all levels of experience, especially considering the decreased operative volume impacting the training of many fellows, residents, and students. Many programs successfully employed virtual curricula, remote simulation activities, and telesimulation to achieve their learning objectives.

Quaranto and colleagues described their successful initiative to utilize increasingly ubiquitous video-conferencing equipment to deliver basic suturing and knot-tying simulation training to surgery clerkship students who were at home during June and July 2020, early in the pandemic.²² They used their institution's video-conferencing vendor, Zoom. In addition to the usual third-person view from a laptop or webcam, they added a high-resolution camera mounted from the instructor's first-person view to allow optimal demonstration of the tasks. They also utilized a cost-effective, slightly modified, open source, 3D printed phone dock to allow each student to share their first-person view with the instructor during the simulation sessions. Notably, they reported a minimal upload bandwidth of 4 Mbps and latency of 70 msec to maintain the video transmission during the session. Each student received a standardized surgical equipment pack that included the 3D printed phone mount, a suture practice pad, a needle driver, forceps, scissors, a 2-color shoelace, two 12-packs of 2-0 silk ties, two 2-0 silk sutures on an SH needle, two 4-0 Prolene sutures, and two 4-0 Monocryl sutures. The course consisted of 3 sessions—an introduction to remote surgical skills simulation and review of the instruments, a session focused on knot tying where students tied knots on the handle of a mug, and a final session focused on suturing. In each session, the instructor first demonstrated the technique and then answered questions and provided feedback, including specific instruction for individual students using the first-person camera dock. The sessions were deemed successful as all students were reported to have visually demonstrated completion of the knot tying and suturing tasks. Students aggregate confidence scores improved from presurvey to postsurvey for knot tying and suturing. Finally, qualitative feedback from the students highlighted the effectiveness of the first-person camera view for the instructors and trainees for a remote simulation session.

Tellez and colleagues reported their experience with the addition of virtual instruction of basic knot tying and suturing tasks for an intern boot camp in the setting of restrictions imposed by the COVID-19 pandemic in June 2020.²³ They compared the 2020 cohort to the 2019 cohort, which participated in a traditional 1-week in-person boot camp. Both cohorts received instruction on 12 tasks (Table 1), had opportunities for voluntary one-on-one tutoring, completed testing at the end of the boot camp period, and received ongoing one-on-one coaching until achieving proficiency in the tasks. The 2019 in-person cohort was introduced to the skills during a

Table 1
Intern curriculum tasks.

Task	Description
1	Palm needle driver
2	Knot tying, No tension, 2-Handed
3	Knot tying, No Tension, 1-Handed
4	Knot tying, Under tension, 2-Handed, Surgeon's knot
5	Knot tying, Under tension, 2-Handed, Slip knot
6	Knot tying, Under tension, 1-Handed, Slip knot
7	Suturing, Interrupted, Simple
8	Suturing, Interrupted, Horizontal mattress
9	Suturing, Interrupted, Vertical mattress
10	Suturing, Running, Simple
11	Suturing, Running, Subcuticular
12	Suturing, Interrupted, Subcuticular

1-hour session at the beginning of the boot camp week. They were provided with instructional videos of the tasks previously recorded by Scott and colleagues, who validated the curriculum. Over the week, they had 4 more hours of practice with faculty, access to the simulation center for independent practice, and voluntary one-on-one tutoring sessions with the faculty simulation director. However, no interns participated in the one-on-one tutoring. Due to pandemic related restrictions on in-person activities, the 2020 boot camp was reduced to one half of the 2019 sessions and additional sessions were conducted via Zoom over the entire month. The incoming interns received knot tying and suturing practice materials at home. The curriculum was introduced in 3 one-hour sessions, and participants were given instructional videos, newly recorded since the prior year, after each session. Three virtual tutoring sessions with a 1:4 faculty-to-intern ratio were held before the truncated in-person boot camp. The interns were again offered one-on-one tutoring during the boot camp period, and 9 interns accepted the offer for in-person, socially distanced tutoring while 9 interns accepted the offer for virtual tutoring.

The performance of the 2020 cohort was markedly improved compared to the 2019 cohort. In 2019, no residents achieved proficiency in all tasks at the boot camp post-test, and only 5.7% of tasks tested across all interns (18 out of 312) were scored as proficient. In contrast, in 2020, 8 residents (27.6%) attained proficiency in all tasks and 86.8% (302 out of 348) of tasks tested across all interns were scored as proficient. The 2020 cohort demonstrated statistically significant decreased time to completion and increased score for all tasks compared to the 2019 cohort. Finally, in 2019 the faculty simulation director spent 55.5 hours providing one-on-one coaching with interns after the posttest, resulting in 64% of tasks rated as proficient by November. In 2020, the director spent 18 hours providing one-on-one coaching after the post-test, resulting in 92% of tasks rated as proficient by September. The authors note that the virtual format was effective in helping the interns achieve proficiency in the curriculum's tasks. However, they acknowledge that other factors, including the provision of practice materials at home, extension of the practice period in the month before clinical duties began, and new instructional videos, likely contributed to improved performance in 2020. Additionally, they report the possibility of a "COVID effect," in which limitations on travel and in-person gatherings may have resulted in more time devoted to training compared to other activities in the month before starting the intern year. Regardless, given the marked improvement in performance, the authors planned to continue the 2020 curriculum with preboot camp virtual sessions in 2021 and likely beyond.

Chan and colleagues reported development of a virtual technical skills course for senior lung transplantation fellows to attempt to minimize skill decay during the clinical slowdown at the start of the pandemic.²⁴ The Toronto Lung Transplant Program suspended transplantation activity, except for a select few acutely ill patients, for months during the early period of the COVID-19 pandemic. They ultimately performed only 12 lung transplants in a period that saw 50 lung transplants in 2019. During that period, 7 fellows were provided a set of equipment and supplies at home that included a long-shafted Castroviejo needle driver, forceps, a novel stand

with clips to mount the simulated tissue, rubber balloons to simulate the thin-walled pulmonary artery and left atrium, and 5-0 Prolene sutures. Teaching sessions were held twice weekly for 3 months via Microsoft Teams video conferencing platform. The model could be adapted to simulate various technical challenges encountered during lung transplantation, including donor-recipient vessel size mismatch, short vessel cuffs, and specific suture methods for the left atrial cuff. During each session, volunteer fellows performed an anastomosis on camera with real time faculty commentary on their technique and advice on dealing with challenges in the real clinical setting. Each fellow also recorded an attempt at the anastomosis on a mobile phone, which was sent to a staff surgeon for a private critique with advice on deficiencies and suggestions for improvement. With permission of the individual fellows, some videos were also shown at the teleconference for group review. In addition to the virtual technical skills sessions, the fellows also had a series of virtual didactic and discussion sessions focused on practical aspects of lung transplantation.

Twelve total sessions were held during the 3-month period. Participating fellows had an average of 13.9 years from medical school graduation before starting lung transplant fellowship and had performed an average of 25.9 lung transplant cases before the pandemic. Fellows reported spending an average of 5 hours (range, 1.3-15 hours) practicing at home. Five out of 7 fellows reported self-described improvement in technical skills after participation in the curriculum and 5 out of 7 reported improved confidence in performing a lung transplant when clinical activities resumed. Feedback from the fellows was mostly positive, including reporting that review of individual videos was useful, that having equipment for independent practice was valuable, and that the regular meetings provided motivation for continued practice. The fellows felt that the simulation model was not particularly realistic, particularly noting the difference in behavior of the balloon and an actual pulmonary artery, but they still felt the model was useful. The program evaluated the warm ischemic time for participating fellows' 5 most recent cases before the pandemic and first 5 cases on return from the pandemic. Among all fellows, they found no statistically significant difference in warm ischemic time after the pandemic, although 1 fellow had an individual statistically significant improvement in warm ischemic time, from 86.6 minutes to 63 minutes ($P = 0.05$). Overall, the authors found that the curriculum of virtual technical skills teaching, independent practice with a simulation model, individual video review, and virtual discussions helped senior trainees maintain their technical skills in a complex procedure during an unexpected cessation of clinical activities.

Greenberg and colleagues adapted an in-person hands-on training course on incisional hernia repair for practicing surgeons to a virtual at-home format, the Hernia@Home course, during the pandemic.²⁵ The virtual course was part of SAGES year-long ADOPT (Acquisition of Data for Outcomes and Procedure Transfer) method, replacing the in-person course that normally would have taken place at the SAGES annual meeting. Aside from the conversion to a virtual format for the hands-on portion of the course, the remainder of the course followed SAGES' usual ADOPT format, including expert faculty trained in the Laparoscopic Colectomy Train the Trainer (Lapco TT) course, precourse lectures and videos, a precourse virtual meeting to set learning objectives, and monthly webinars for a year after the virtual hands-on session. The course organizers chose an existing telementoring platform from Proximie Ltd, which uses an augmented reality interface allowing real-time shared viewing of the camera view and telestration from the mentor. The organizers planned to use a human cadaveric model in the in-person hands-on course to teach tissue-based inguinal hernia repair and anterior and posterior component separation techniques for ventral hernia repair. With conversion to a virtual format, they chose to focus on component separation for ventral hernia repair and worked with KindHeart, a company that builds porcine surgical models, to develop a porcine abdominal wall model and tailored trainer box to house the model. The custom trainer box, mesh, and suture were shipped to participants in the week before the course. The porcine abdominal wall model, which was vacuum sealed in preservative and frozen, was shipped to participants 2 days before the course. Learners supplied their own surgical instruments. On the day of the course, learners set up their own trainer boxes, abdominal wall models, and webcams. The virtual course was conducted in trios of 1 faculty and 2 learners, with each learner having a chance to be virtually mentored in the operative procedures

via audible and telestrated feedback from the faculty and a chance to observe the other learner's operative session.

Fifteen learners and 8 faculty members participated in the virtual hands-on session, which lasted from 38 to 190 minutes. A majority of course participants were able to set up the model and webcam without difficulty, but 2 surgeons required assistance from Proximie personnel on the day of the session. Logistical challenges reported by the participants included lighting and camera positioning (21%), issues logging into the platform (21%), difficulty coordinating schedules due to clinical responsibilities (21%), and difficulty obtaining instruments (7%). All attendees felt the porcine model was valuable for the learning experience and all but one felt the telementoring platform was valuable. All attendees reported that the virtual format allowed them to achieve their personal learning objectives and that the session was relevant to their professional development. Ten of the course participants had previously participated in in-person courses. They felt the virtual format was similar or even offered advantages compared to an in-person course, including lack of travel, fewer participants per model, and increased time with the faculty mentor. The organizers noted a few drawbacks, including that the virtual course was expensive due to the cost of developing and shipping the operative model, that they were unable to ship models to participants outside the United States, and that the course required more logistic coordination than a course offered at a single site. They believed their model for a successful virtual hands-on course could promote continued operative professional development during the pandemic and could persist beyond the pandemic to allow training options for surgeons with travel and time restrictions in the future.

Surgical simulation changed and adapted in significant ways during the COVID-19 pandemic, many of which will carry forward into the future. This was exemplified by a panel discussion at the 2024 Association of Surgical Education annual meeting titled "Remote Skills Training: It's Here to Stay." Although surgical simulation fundamentally changed in many ways during the pandemic, it remains a core element of surgical training, even taking on increased importance in light of the loss of operative experience by trainees. The special committee of the ACS Academy of Master Surgeon Educators conducted a follow-up survey of leaders in surgical education on the impact of the COVID-19 pandemic that was distributed between June and August 2021.²⁶ Overall, although programs experienced lingering effects of the pandemic on surgical education, they reported less disruption of educational activities, restoration of clinical activity, adaptation of curriculum delivery, and improvement in trainee well-being compared to 2020. Almost all programs had transitioned educational conferences to a virtual format, but only a small percentage transitioned simulation to a virtual format and only approximately 40% reported in-person simulation activities during the 2020-2021 academic year. However, leaders noted that increasing simulation would be a key adjustment to remediate deficiencies in minimal case numbers and progression of operative autonomy.

Use of simulation to respond to COVID-19

Although routine surgical simulation activities and assessment slowed or stopped early in the pandemic, simulation resources were redeployed in many institutions to help understand and respond to the rapidly changing clinical environment. At one of the author's institutions, the leadership of a recurring combined surgical and anesthesia in situ simulation program utilized that infrastructure to conduct simulations of patient transport and personal protective equipment use through the perioperative spectrum for COVID positive patients requiring emergency surgery. Orgill and colleagues described a "pop up" simulation that was developed and implemented in 2 days to teach proper use of personal protective equipment, safe transport, and minimizing aerosols during intubation and extubation for anesthesia clinicians and technicians.²⁷ New simulation models were developed to address the unique challenges of the pandemic. Gomez and colleagues described their development of a high-fidelity aerosol-generating cough simulator²⁸ (video link, <https://www.youtube.com/watch?v=7FEDFz1NmBU>, (Fig 4).



Fig. 4. Video link to coronavirus (COVID) model.

They used polyvinyl chloride (PVC) pipe, a bicycle pump valve, adapters, and a ball valve connected to airway circuit tubing. The tubing was connected to an endotracheal tube placed retrograde through the bronchus of a mannequin. They used a UV-visible product to coat the mannequin's airway to simulate production of aerosolized particles. The total cost of the model was \$108. Kravitz and colleagues developed a virtual reality (VR) training module for donning and doffing of personal protective equipment using the Oculus Quest that was administered from November 2020 to January 2021.²⁹ Participants who used the VR training module performed better, although not achieving statistical significance, than a comparison group that completed an e-module with an instructional video and PowerPoint presentation. Creative utilization and rapid adaptation of preexisting simulation resources, as occurred in the COVID-19 pandemic, can be critical elements in the response to any new clinical challenge that may emerge in the future.

The evolving landscape of surgical education

The traditional apprenticeship model in surgical training, where trainees learn by observing and assisting experienced surgeons, has limitations, especially as surgical procedures become more complex. There is a growing need for structured and standardized training methods to ensure patient safety and objectively evaluate trainees' performance. The rise of AI in surgical education has introduced more comprehensive, safe, and efficient learning experiences.³⁰

In a Delphi study done by the ACS, a group of experts stated that consensus that AI methods and AI-enabled metrics have the potential to benefit surgical education, including training (100%), assessment (97.5%), and feedback (100%) was reached. For risks with AI and metrics, the panel only approached consensus on loss of privacy (75%) and the impact of human biases on AI (70%). Notably, there was no consensus that failure to be interpretable (57.5%) and actionable (62.5%) are potential risks with AI for surgical education.³¹

Nontechnical training

Nontechnical skills are increasingly recognized as critical components of surgical competence. These skills include communication, teamwork, leadership, decision-making, situational awareness, and patient note writing. Effective nontechnical training focuses on several areas:

Communication skills. Surgeons must communicate effectively with patients, families, and health care teams. Training programs incorporate role-playing, standardized patient interactions, and simulation-based exercises to enhance verbal and nonverbal communication skills.

Teamwork and collaboration. The OR is a complex environment that requires seamless collaboration among various professionals. Team-based training exercises, including crisis resource management simulations—a set of principles dealing with cognitive and interpersonal behaviors that contribute to optimal team performance—help to foster a culture of teamwork and mutual support.

Leadership and decision-making. Surgeons often lead clinical teams and make critical decisions under pressure. Leadership training programs, mentorship, and scenario-based exercises develop these essential skills.

Situational awareness. Recognizing and responding to changes in the surgical environment is crucial for patient safety. Simulation exercises and debriefing sessions improve situational awareness and adaptive thinking.

Patient notes writing skill. Accurate and comprehensive patient note writing is essential for ensuring continuity of care, effective communication among health care providers, and legal documentation. Training programs emphasize the importance of thorough documentation, including preoperative assessments, operative details, postoperative care, and follow-up notes. Structured templates, feedback from experienced surgeons, and electronic health record (EHR) systems are used to enhance these skills.

Interprofessional and multidisciplinary health professions training

Modern surgical education emphasizes the importance of interprofessional and multidisciplinary training. This approach prepares surgeons to work effectively within diverse health care teams, enhancing patient care and outcomes.

Interprofessional education. Interprofessional education involves collaborative learning experiences where students from different health professions learn with, from, and about each other. This fosters mutual respect and understanding, leading to better team dynamics in clinical practice.

Multidisciplinary team training. Surgical education programs increasingly integrate multidisciplinary team training, involving surgeons, nurses, anesthesiologists, and other health care professionals. Simulated OR scenarios and team-based problem-solving exercises enhance coordination and collaboration.

Shared decision-making: Training programs emphasize the role of shared decision-making in patient care, encouraging surgeons to engage patients and their families in the treatment process. This holistic approach ensures that all perspectives are considered, leading to better-informed and more patient-centered care.

Simulation

Accreditation

American College of Surgeons Accredited Education Institutes

The ACS AEI program began in 2005 and has a stated goal to “promote patient safety by using simulation, to support the lifelong learning of surgeons, surgical trainees, and other healthcare professionals, to enhance access to contemporary surgery education, and to advance scholarship in simulation-based surgical education.” As of this writing, the ACS has accredited 94 comprehensive or focused education institutes across 34 states and the District of Columbia and 10 other countries (Fig 5). Twenty of those sites offer AEI-accredited fellowship programs in simulation. The accreditation program is managed by the ACS Division of Education. The accreditation process is rigorous to ensure accredited sites meet the program’s standards. To achieve



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Fig. 5. Seal of Accredited Education Institutes (AEI).



Fig. 6. First multiuser video trainer at University of Texas Southwestern in Dallas, Texas.

accreditation, prospective sites must complete an application process and site visit. Maintenance of accreditation requires annual reports and a re-accreditation process with site visits every 5 years.

The program's standards, introduced in 2005, were updated in 2012 and again 2021. The standards³² are organized into 5 categories with corresponding criteria. These include personnel, governance and budget, learners and educational resources, educational methodology, and advancement of the field. The personnel requirements ensure the institute has appropriate, qualified leadership with protected time and sufficient administrative and support staff. The governance and budget requirements ensure that the institute has a clear statement of purpose to inform its goals and programs, adequate budget and ongoing financial support, appropriate organizational structure, and an oversight committee that meets at least twice annually. The learners and educational resources criteria ensure that the institute is accredited by an appropriate governing body (ie, Liaison Committee on Medical Education [LCME], ACGME) and provides education to surgeons in practice and at least 3 other learner groups, which may include surgical residents, medical students, nurses, physicians or residents from other disciplines, allied health professionals, and parents and families (Fig 6). The standards require Comprehensive Institutes to have at least 1200 square feet of space controlled by the institute and dedicated to education and training with up to 4000 additional square feet available for education and training needs of learners as well as activities of administrative and support staff. Focused Institutes must have at least 800 contiguous square feet of space dedicated to learner education and training and be able to accommodate at least 6 learners at 1 time (Fig 7).

The institute must maintain an inventory of necessary devices and equipment for their planned curricula and have staff who can appropriately use them. The educational methodology criteria ensure the institutes are developing new curricula using a structured approach, select

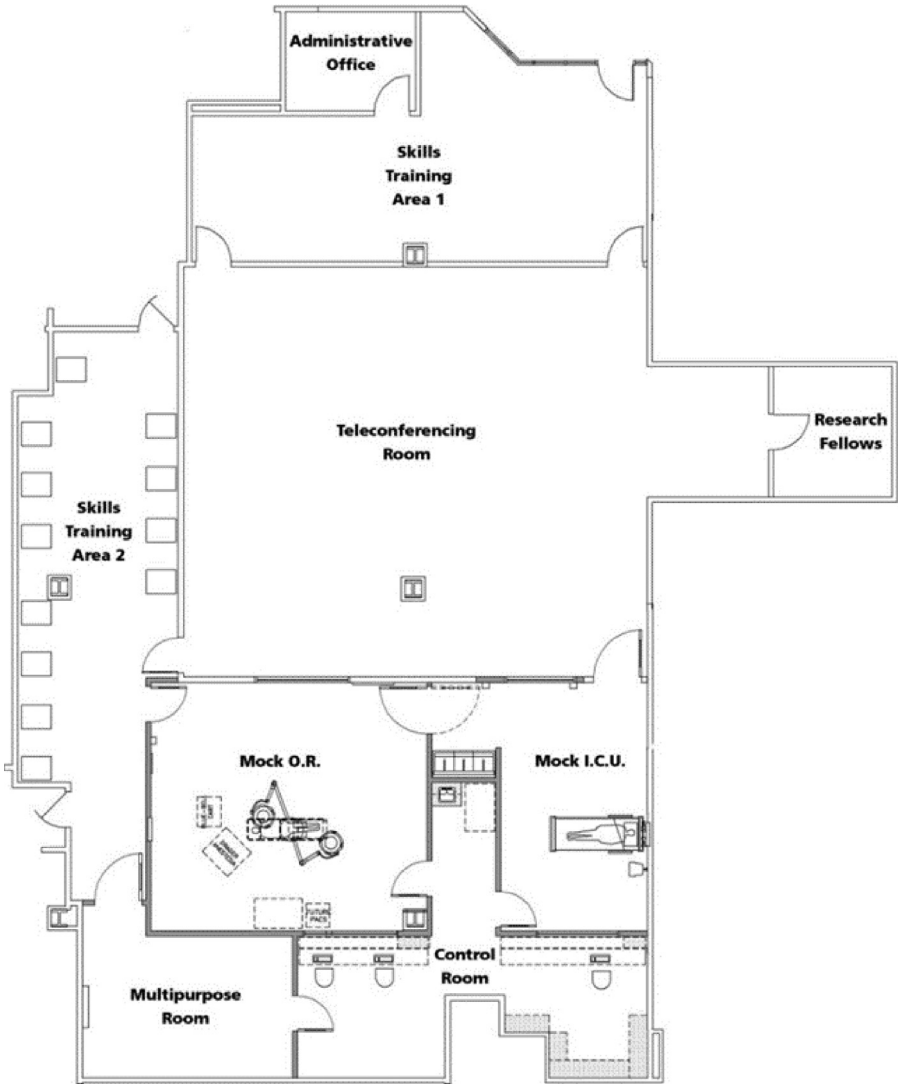


Fig. 7. Floor plan for Carl J. Shapiro Simulation and Skills Center at Harvard University.

appropriate methods and models to meet learning objectives, include cognitive, psychomotor, and affective domains in their scope of education, conduct assessment and evaluation of learners, faculty, and programs, and provide faculty development. Finally, the advancement of the field criteria requires institutes to conduct research or scholarly activities in simulation-based education, submit annual compliance reports, and engage with the consortium of ACS-Accredited Education Institutes. Each criterion in the 5 domains has explicit compliance measures evaluated for initial accreditation and monitored for maintenance of accreditation.

Society for Simulation in Healthcare

The Society for Simulation in Healthcare (SSH) also offers accreditation of simulation centers³³ although without the specific focus on surgical simulation. The SSH accreditation website states that accreditation can improve quality of care, increase visibility, and enhance credibility



Fig. 8. Insignia of the Society of American Gastrointestinal and Endoscopic Surgeons (SAGES) Fundamentals courses: Fundamentals of Laparoscopic Surgery (FLS), Fundamental Use of Surgical Energy (FUSE), and Fundamentals of Endoscopic Surgery (FES).

for accredited programs. Like the ACS-AEI program, the SSH updated its accreditation standards in 2021. All programs seeking accreditation must meet the core standards. The core standards include criteria for mission and governance, program management, resource management, human resources, program improvement, integrity, and expanding the field. Programs must also meet standards in at least one of the areas of assessment, research, and/or teaching/education. Programs may also apply for accreditation in the areas of systems integration and fellowship. The SSH has 2 accreditation cycles per year and successful programs are granted accreditation for 5 years.

Curricula

The most successful and widely disseminated simulation curricula employ a combination of structured didactic materials and tailored simulation activities. Most are developed and implemented by major surgical societies that have the financial and personnel resources to build complex curricula and keep them up to date. The following are some of the successful programs from the SAGES, ACS Division of Education and the Committee on Trauma, the ASE, and the APDS.

Society of American Gastrointestinal and Endoscopic Surgeons (SAGES)

SAGES has been and continues to be a leader in the development, implementation, and wide dissemination of simulation-based surgical training curricula and assessment. They launched the first "Fundamentals" program, FLS, in 2004. FES followed in 2010 and FUSE launched in 2013 (Fig 8).

Fundamentals of Laparoscopic Surgery (FLS)

The FLS program was launched by SAGES in October 2004.³⁴ However, the idea for the course arose much earlier, around 1997, with the leaders of the SAGES Continuing Education Committee, Drs. Lee Swanstrom and Nathaniel Soper. They were inspired to create something akin to the ATLS course to teach core knowledge and skills for laparoscopy, which was becoming widely used but with wide variability in training programs and outcomes. The program was envisioned to include didactics, clinical judgment, and technical skills. For the technical skills portion, Dr. Gerald Fried was engaged to adapt the McGill Inanimate System for Training and Evaluation of Laparoscopic Skills (MISTELS) for the FLS program. The trainer that would become the FLS box was initially developed by Dr. Carl Westcott, who donated the intellectual property. The developers of the program wanted to ensure that the cognitive and technical skills assessments were structured and validated to create a meaningful certification.

For the cognitive examination, a psychometrician was engaged, and beta testing occurred at 8 sites to validate the examination before widespread use. SAGES partnered with the ACS in 2005 and formed a joint committee to oversee the FLS program. As noted previously, in 2008 the ABS mandated that all general surgery residents obtain FLS certification to be eligible for the ABS qualifying examination. The ABOG implemented an FLS certification requirement for new applicants in 2020.

Table 2
Fundamentals of Laparoscopic Surgery (FLS) modules.

0	Introduction
1	Laparoscopic equipment
2	Energy sources
3	Operating room setup
4	Patient selection/preoperative assessment
5	Anesthesia and patient positioning
6	Pneumoperitoneum establishment and trocar placement
7	Physiology of pneumoperitoneum
8	Exiting the abdomen
9	Current and diagnostic laparoscopic procedures
10	Biopsy
11	Laparoscopic suturing
12	Hemorrhage and hemostasis
13	Postoperative care and complications
14	Manual skills training
	final review

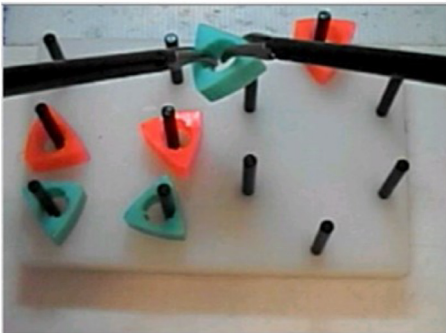


Fig. 9. Fundamentals of Laparoscopic Surgery (FLS) peg trainer task.

The FLS curriculum is available via the SAGES online platform at <https://www.sages.org/owls/>, in both English and Spanish. The comprehensive web-based didactic modules cover a range of subjects related to laparoscopy, shown in Table 2. The modules also include detailed videos of the tasks required for the technical skills portion of the FLS examination. The proctored FLS assessment includes a computer-based, timed, multiple-choice examination and a manual skills evaluation. The cognitive examination tests understanding of the basic principles of laparoscopy, focused on clinical application. The manual skills evaluation includes 5 tasks with task-specific completion criteria. The tasks include peg transfer (Fig 9), precision cutting, ligating loop, suturing with extracorporeal knot tying, and suturing with intracorporeal knot tying. Candidates achieving a passing score on the cognitive examination and manual skills examination are granted certification.

The value of FLS certification has been recognized for both trainees and surgeons in practice. A recent study by Higgins, Turbati, and Goldblatt matched data from the national FLS registry with Society for Improving Medical Professional Learning (SIMPL) evaluations of general surgery resident intraoperative performance from 2015 to 2021.³⁵ They found that residents required less supervision and had improved performance scores in laparoscopic operations performed after they passed the FLS examination compared to operations performed before they passed the FLS examination. In 2005, the Controlled Risk Insurance Company of Harvard’s Risk Management Foundation (CRICO/RMF), the internal malpractice underwriter for 18 teaching hospitals, sponsored FLS training for laparoscopic surgeons in practice and provided a malpractice premium

rebate for those achieving certification.³⁶ The joint SAGES and ACS committee for FLS issued a statement in 2012 recommending that all surgeons performing laparoscopy be certified in FLS.

Telesimulation and teleproctoring

Telesimulation, the use of telecommunication resources to conduct real-time simulation activities when the trainer and trainee are geographically separated, was first described in 2010 by Drs. Okrainec, Henaoui, and Azzie, a group of surgeons from the University of Toronto.³⁷ Surgeons from the University of Toronto had traveled to Botswana in March 2007 to teach a 3-day in-person FLS course. Although the course participants showed notable improvement in the assessed laparoscopic skills after the course, only 10% of participants achieved FLS certification. The concept of telesimulation arose from the perceived need for further guided practice for the surgeons in Botswana in the face of costly and time-consuming travel for the surgeons from Toronto. Dr. Okrainec and his colleagues developed a telesimulation setup in Botswana and Toronto and conducted a controlled trial to evaluate the effectiveness of teaching FLS skills using telesimulation.

Dedicated telesimulation rooms were set up at the University of Botswana and University of Toronto. The setup included a laparoscopic trainer box with a camera connected directly to a display screen. The camera also connected to a laptop, allowing its use as a webcam. A second webcam at each site recorded an external view of the surgeon. The webcams and surgeons were connected by Skype. At each location, the instructor or trainee had a screen showing their own laparoscopic trainer box, a screen showing the collaborating surgeon's trainer box, and a projected image of the collaborating surgeon. The instructor and trainee communicated via laptop microphones and speakers.

The participants in the study included 16 surgeons from Botswana, 13 practicing surgeons and 3 trainees. The group was split into 8 surgeons in a telesimulation group and 8 surgeons in a self-practice group. Each group received an FLS instructional DVD and a FLS simulator with the necessary instruments and supplies for the 5 FLS tasks: peg transfer, pattern cutting, ligating loop, extracorporeal knot tying, and intracorporeal knot tying. All participants were told to practice the FLS tasks for at least 2 hours per week for 8 weeks. The surgeons in the telesimulation group had a weekly, 15-minute telesimulation session with a surgeon from Toronto. During those sessions, the surgeons in Botswana demonstrated as many tasks as possible and received focused instruction and demonstration of technique from the surgeon in Toronto. The sessions were scheduled at the end of the clinical day in Botswana, which was the morning in Toronto. After the 8 weeks of practice, 2 surgeons from Toronto traveled to Botswana for in-person, official FLS certification testing. The surgeons in the telesimulation group achieved significantly higher scores on all FLS tasks except the ligating loop, which had higher scores in the telesimulation group but did not achieve statistical significance. All 8 of the surgeons in the telesimulation group achieved FLS certification at that time, compared to only 3 surgeons in the self-practice group.

After the initial proof-of-concept with the FLS telesimulation project in Botswana, the Toronto group demonstrated the efficacy of telesimulation at other sites, for other technical skills, and in collaboration with other specialties. In 2011, they partnered with pediatricians from Toronto to use telesimulation to teach intraosseous needle insertion for pediatric resuscitation to physicians in Botswana.³⁸ In 2013, they published a report of their successful efforts to teach FLS skills to surgeons in Colombia using telesimulation.³⁹ In 2016, they used their telesimulation platform to teach ultrasound-guided regional anesthesia to anesthesiologists at 10 hospitals in Ontario, Canada.⁴⁰ To expand on their use of telesimulation instruction, they demonstrated the feasibility of remote assessment to promote structured feedback in telesimulation⁴¹ and for formal certification in the FLS program.⁴²

International Dissemination of Fundamentals of laparoscopic surgery (FLS) and the Global Laparoscopic Advancement Program

SAGES has extended their reach internationally through translation of the FLS curriculum, expansion of training and testing sites, and recently, a new program called the Global Laparo-

Table 3
Fundamentals of Endoscopic Surgery (FES) modules.

1	Technology
2	Patient preparation
3	Sedation and analgesia
4	Upper gastrointestinal endoscopy
5	Lower gastrointestinal endoscopy
6	Performing lower GI procedures
7	Small bowel
8	Diagnostic endoscopic retrograde cholangiopancreatography
9	Hemostasis
10	Tissue removal
11	Enteral access
12	Endoscopic ultrasound
13	Advanced imaging

scopic Advancement Program (GLAP).⁴³ The GLAP involves a collaboration between SAGES and a national surgical society or local government in another country. SAGES has a team of faculty experts which partners with local surgeon educators to teach the curriculum. The GLAP course includes 5 days of instruction on “train the trainer” techniques, laparoscopic skills, curriculum development, and outcomes and data reporting. Local educators and trainees participating in the course must complete virtual precourse education modules. Each site must develop a plan for GLAP implementation, including buy in from local hospitals, sustainability, timeline, and assigned responsibilities. In addition to lectures, course participants spend 2 days operating with visiting faculty and local educators develop their own low-cost simulators. The program is intended to be a long-term, sustainable partnership between the local site and SAGES, with ongoing data collection, outcomes reporting, mentorship, and progress reports. Tsai and colleagues reported their successful development of an ambulatory surgical center in Uganda via a partnership with a Ugandan surgeon and the Mount Sinai Department of Surgery in New York.⁴⁴ The local surgeon completed a minimally invasive surgery mini-fellowship and trained with the FLS program. The team then completed a weeklong in-person training with the Ugandan OR team, after which the surgeon performed laparoscopic cases with telementoring from the surgeons in the United States. From October 2021 to February 2024, the surgeon performed 37 laparoscopic appendectomies and 24 laparoscopic cholecystectomies with telementoring, with no intraoperative complications and only 2 superficial wound infections noted on short-term follow-up. The GLAP aims to produce surgeons who can safely and independently perform laparoscopic surgery, create an enduring educational environment at the partner site, develop ongoing data collection and outcomes reporting to promote patient safety and quality, and empower host countries to independently maintain ongoing training programs with local faculty after the partnership with SAGES.

Fundamentals of Endoscopic Surgery (FES)

The first discussions about a standardized endoscopy curriculum were held at the 2005 SAGES meeting, after which the FLS task force was created.⁴⁵ The task force had 2 subcommittees, 1 focused on the developing program’s didactic materials and 1 on creation of a platform for testing manual endoscopic skills. The program was officially launched in 2010. As with FLS, the program’s didactic materials are available via the SAGES organization wide learning system online. There are 13 didactic modules, outlined in [Table 3](#). The FES assessment includes a 90-minute computer-based multiple-choice examination and a manual skills test to document adequate technical and psychomotor skills. The skills test is performed on a GI Mentor II, GI Bronch, GI Express, or Endo Mentor Suite Virtual Reality Trainer at designated test centers ([Fig 10](#)).

The proprietary FES exam modules evaluate the examinee’s efficiency and precision on 5 simulation exercises: navigation, loop reduction, retroflexion, mucosal evaluation, and targeting/tool manipulation. Both portions of the examination are taken in the same testing appoint-



Fig. 10. GI Mentor for a Fundamentals of Endoscopic Surgery (FES) examination.

ment. The ABS incorporated the FES Assessment as a final milestone of their Flexible Endoscopy Curriculum (FEC), which is now required of all general surgery candidates for board certification. The FEC outlines the recommended foundational knowledge and endoscopy experience, obtained via simulation and through minimum required cases, for general surgery residents. Given the mandatory FES certification requirement from the ABS and ACGME case minimums for endoscopy, endoscopic skills training, commonly using the FES curriculum, is now incorporated in all ACGME-accredited general surgery training programs.

Fundamental Use of Surgical Energy (FUSE)

The FUSE program was conceived by SAGES president-elect Dr. Steven Schwaitzberg in 2009 in response to a perceived deficit in understanding of energy devices used every day in surgery. In 2010, the SAGES president, Dr. Jo Buyske, created the FUSE task force, initially led by Drs. Daniel Jones, Liane Feldman, and Pascal Fuchshuber. From the outset, FUSE was envisioned as a universal educational program for any physician, nurse, or allied health professional who is involved in the use of energy devices in the clinical setting.⁴⁶ A pilot FUSE course was held at the SAGES meeting in 2011. At that time, an 11-item multiple choice pretest covering critical knowledge of surgical energy was administered to 48 SAGES leaders.⁴⁷ The median percentage correct was 59% and 31% of respondents gave incorrect answers regarding management of an operating room fire and device interference with cardiac pacemakers. The same test was administered to FUSE course participants, whose median percent correct increased from 55% before the course to 90% after the course, providing further support for the need for this critical curriculum. A SAGES manual on the FUSE was developed and published in 2012. The manual content was subsequently transformed to an online modular curriculum, with module topics outlined in [Table 4](#). The online curriculum is available free through the SAGES online organization wide learning system. The FUSE didactic modules are available in English, French, Spanish, and Chinese. FUSE has also developed hospital compliance modules for use in institutional training. As with the other Fundamentals programs, the FUSE program offers a certification to those who successfully complete a validated, multiple-choice certifying examination.

Although the FUSE didactic curriculum does not include hands-on simulation, the content has been adapted for use in a wide range of hands-on simulation activities. A Virtual Electrosurgery Skill Trainer (VEST) was developed by a group led by Suvranu De. In an initial validation study, participants felt the VR model was effective for learning electrosurgery fundamentals and almost one half of participants preferred the VEST simulator compared to an animal model.⁴⁸ Madani and colleagues describe a 2-hour FUSE curriculum consisting of a 1-hour didactic course on electrosurgical energy followed by a 1-hour goal-directed hands-on simulation session.⁴⁹ The

Table 4
Fundamental use of surgical energy (FUSE) modules.

0	FUSE introduction
1	Fundamentals of electrosurgery
2	Mechanisms and prevention of adverse events
3	Monopolar devices
4	Bipolar devices
5	Radiofrequency for soft tissue ablation
6	Endoscopic devices
7	Ultrasonic energy devices
8	Microwave energy systems
9	Energy devices in pediatric surgery
10	Integration of energy systems with other devices
11	Surgical smoke and aerosolization safety

simulation course was delivered to trainees at 11 residency programs in Canada, the United States, and the United Kingdom. Residents were randomized to complete the didactic session followed by the structured simulation session versus an unstructured session where trainees could use electrosurgical devices. All participants improved their score on the examination after the intervention, but with higher scores in the structured simulation group. The simulation group also had higher scores 3 months after the session, although all scores decreased at the 3-month follow-up. The investigators followed up with trainee participants 1 year after the course. They found that trainees who completed the structured simulation curriculum maintained scores significantly above their pretraining baseline, whereas those in the unstructured practice control group had scores that were similar to their baseline.⁵⁰ Altieri and colleagues subsequently demonstrated that the use of telesimulation for the structured electrosurgical simulation was as effective in improving trainees' knowledge and comfort in the safe use of electrosurgical devices as an in-person session delivered by an on-site proctor.⁵¹

Society of American Gastrointestinal and Endoscopic Surgeons (SAGES) Masters Program

After fellowship, it is challenging for surgeons in practice to develop new technical skills. SAGES created a structure, the SAGES Masters Program, to help surgeons learn new operations in a graded pathway from proficiency, to competency, to master.⁵² The curricula includes Fundamentals and relies on simulation and coaching.

Collaborative curricula from the American College of Surgeons, Association for Surgical Education, and Association of Program Directors in Surgery

The ACS Division of Education has partnered with the ASE and APDS to develop, disseminate, and maintain surgical skills curricula that span the entire 4 years of medical school, the transition to surgical residency, and the 5 clinical years of a general surgery residency.

American College of Surgeons (ACS)/Association for Surgical Education (ASE) medical student simulation-based surgical skills curriculum

The ACS and ASE Medical Student Simulation-Based Surgical Skills Curriculum originated in ASE at the 2009 Surgical Education Week.⁵³ The impetus for the curriculum arose from the observation that, due to a range of factors, students had diminishing opportunities for direct patient care in medical school but were still expected to perform at the same level on day 1 of residency. A needs analysis of 4 medical schools identified that medical students sought simulation in the first 2 years of medical school, while the clerkship directors were more inclined to wait until the third and fourth year of medical school before introducing simulation.⁵⁴ The ACS Division of Education joined the effort in 2010.

The ACS ASE Steering Committee was headed by Drs. Kathy Liscum and Dan Jones. An initial list of topics was developed by expert consensus. The joint committee of the ACS and ASE subsequently conducted a needs assessment survey of clerkship directors nationally and fourth year medical students from 5 schools in distinct geographic regions to hone the final list of topics.⁵⁴ The curriculum's modules were developed and edited from 2010 to 2012 (Table 5). The modules

Table 5
American College of Surgeons/Association for Surgical Education medical student simulation-based surgical skills curriculum.

Year 1	Module 1:	Acute bleeding–packing and tourniquet
	Module 2:	Abdominal exam
	Module 3:	Basic vascular exam
	Module 4:	Breast exam
	Module 5:	Digital rectal exam
	Module 6:	Female pelvic exam
	Module 7:	Male groin and genital exam
	Module 8:	Universal precautions
	Module 9:	Venipuncture and peripheral IV
Year 2	Module 1:	Basic airway management
	Module 2:	Communication–history and physical and case presentation
	Module 3:	Foley bladder catheterization
	Module 4:	Intermediate vascular exam
	Module 5:	Nasogastric tubes
	Module 6:	Sterile technique–gloving and gowning
	Module 7:	Surgical drains–care and removal
	Module 8:	Wounds–basic principles
Year 3	Module 1:	Arterial puncture and blood gas
	Module 2:	Basic gastric tube placement and care
	Module 3:	Basic knot tying
	Module 4:	Basic splinting – a universal physician's skill
	Module 5:	Basic stoma care
	Module 6:	Basic suturing
	Module 7:	Central venous line insertion
	Module 8:	Communication–during codes and safe and effective handoffs
	Module 9:	Intermediate airway
	Module 10:	Intraosseous IV
	Module 11:	Local anesthetics
	Module 12:	Lumps and bumps
	Module 13:	Paracentesis
	Module 14:	Thoracentesis
	Module 15:	Ultrasound basics
Year 4	Module 1:	Organ-specific ultrasound diagnosis

are designed to teach core physician skills for all medical students, regardless of intended specialty. Each module includes an outline of the intended simulation activities and an assessment rubric. The assessments are designed for formative feedback to students, not as a summative evaluation. Many modules included supplemental materials including instructional videos. The modules were designed to progress from year to year of medical school, but given the modular nature, they can be adapted to fit any institutional curriculum. The curriculum was released to ASE members in 2013 and nationally in 2014. The curriculum is available at no cost to students and faculty who register with a valid email address. Early implementation of select program modules across multiple medical schools showed that nearly all learners achieved proficiency in suturing, knot tying, and basic airway skills.^{55,56} An expert review of the ACS/ASE Curriculum achieved consensus on 5 core Entrustable Professional Activities (EPAs) from the American Association of Medical Colleges (AAMC) that could be taught and assessed using the curriculum's modules: gather a history and perform a physical examination, provide an oral presentation of a clinical encounter, give or receive patient handover to transition care responsibility, collaborate as a member of an interprofessional team, and perform procedures.⁵⁷

American College of Surgeons (ACS)/Association of Program Directors in Surgery (APDS)/Association for Surgical Education (ASE) resident preparatory curriculum

The ACS/APDS/ASE Entering Surgery Resident Preparatory Task Force conducted a survey in 2009 of surgery interns to assess their preparedness for essential skills for safe patient care

and common intern tasks.⁵⁸ Interns overall reported feeling moderately prepared for medical knowledge and patient care requirements, but well prepared for professionalism, communication, practice-based learning, and systems-based practice requirements. Interns who had participated in a residency preparatory curriculum during their final year of medical school had statistically significantly higher self-assessed evaluation of their technical skills, professionalism, communication skills, and overall preparation. Ultimately, the task force developed a surgery resident prep curriculum, which was piloted by 39 institutions in 2014. The content focuses on critical skills for the first weeks of residency. The program's goals and objectives are subdivided in the domains of patient care and medical knowledge, technical skills, professionalism, interpersonal skills and communication, practice-based learning and improvement, and systems-based practice.⁵⁹ As of 2023, the ACS reported 133 participating institutions. Those participating institutions have access to content for faculty and students, ready-to-use teaching materials, modular content, and a site to track student progress and produce completion certificates. Programs must deliver the designated essential modules and have access to other modules to use as needed. The curriculum focuses heavily on simulation for core intern tasks, technical skills, communication, and professionalism. The modular content can be adapted to each institution's needs and resources. One interesting essential module is a mock page program administered through Southern Illinois University. The mock page program was reported by Boehler and colleagues.⁶⁰ In 2013 and 2014, they delivered the mock page cases to 213 medical students from 12 medical schools. Participating students could receive a mock call from a standardized nurse or patient anytime during the day or night during the program period. Immediate feedback was provided to the student after the call, and course directors received a weekly debriefing report through the 4 weeks of the program. Students showed improved communication and clinical decision making over the 4 weeks of the course. A 2018 survey of general surgery residency program directors assessed the value of a residency prep curriculum. Based on program director perception, interns who had completed a preparatory curriculum had a statistically significant improvement in technical skills and a nonsignificant trend toward improvement for interpersonal skills, medical knowledge, patient care, systems-based practice, problem-based learning, and professionalism, compared to their peers who did not have a residency preparatory curriculum.⁶¹

American College of Surgeons (ACS)/Association of Program Directors in Surgery (APDS) resident skills curriculum

A Surgical Skills Curriculum Task Force was established in 2005 by the ACS and APDS to develop a standardized simulation curriculum for general surgery residents (Table 6). The committee was led by Gary Dunnington, MD, Helen MacRae, MD, and Debra DaRosa, PhD, who recruited content experts for the proposed modules. The curriculum rolled out in 2007 (phase 1) and 2008 (phases 2 and 3). Phase 1, Core Skills, includes 16 modules intended for residents early in training. Phase 2, Advanced Procedures, include 15 modules intended for mid- to senior-level residents with instructions for simulating common surgical procedures. Phase 3, Team-Based Skills, includes 10 modules designed to teach surgery residents optimal team behavior via simulated patient care scenarios.

The curriculum materials are available free of charge to anyone who registers through the ACS Course Catalog and can be accessed via the links on the overview website - <https://www.facs.org/for-medical-professionals/education/programs/acsapds-surgery-resident-skills-curriculum/>. However, although the course materials are free to review, the cost of full implementation could be significant. A 2011 report from the University of Pennsylvania detailed the costs associated with their implementation and continued delivery of the ACS/APDS Surgical Skills Curriculum. They described a \$4.204 million initial capital expenditure to develop the required simulation space and purchase necessary equipment and a cost of \$12,516 per resident for the yearly 4-week dedicated simulation rotation in the first and second year in which the curriculum is delivered.⁶² A 2010 report from Northwestern University detailed the costs associated with implementation of 14 of 16 phase 1 modules. That study detailed a budget of

Table 6
American College of Surgeons/Association of Program Directors in surgery resident skills curriculum modules.

Phase 1: core skills	Phase 2: advanced procedures	Phase 3: team-based skills
1. Asepsis and instrument Identification	1. Laparoscopic ventral hernia repair	1. Teamwork in the trauma bay
2. Knot tying	2. Open colon resection, lap right colon resection	2. Postoperative hypotension
3. Suturing	3. Laparoscopic sigmoid resection	3. Laparoscopic crisis
4. Wound closure and skin flaps	4. Open right colon resection	4. The preoperative briefing
5. Skin grafts	5. Laparoscopic/open bile duct exploration	5. Laparoscopic troubleshooting
6. Urethral catheterization	6. Laparoscopic ventral/incisional hernia repair (porcine model)	6. Postoperative pulmonary embolus
7. Airway management	7. Laparoscopic appendectomy	7. Postoperative MI (cardiogenic shock)
8. Chest tube insertion	8. Laparoscopic nissen fundoplication	8. Latex allergy anaphylaxis
9. Central line insertion	9. Sentinel node biopsy & axillary lymph node dissection	9. Patient handoff
10. Surgical biopsy	10. Open inguinal/femoral hernia repair	10. Retained sponge on postop chest x-ray
11. Laparotomy opening and closure	11. Laparoscopic inguinal hernia repair	
12. Basic laparoscopy skills	12. Laparoscopic/open splenectomy	
13. Advanced laparoscopy skills	13. Laparoscopic/open cholecystectomy	
14. Hand-sewn bowel anastomosis	14. Gastric resection and peptic ulcer disease	
15. Stapled bowel anastomosis	15. Parathyroidectomy/Thyroidectomy	
16. Arterial anastomosis		

Table 7
Promulgation of selected American College of Surgeons Committee on Trauma educational course offerings.⁶⁵

	ATLS 2004-2021	ATOM 2004-2020	ASSET 2008-2021	BEST 2016-2021
Total courses	51,427	1214	1043	144
Total students trained	779,847	5829	6492	817
Total countries	86	16	39	25
Total US states/Canadian provinces	All	28	39	25

ATLS, advanced trauma life support; ATOM, advanced trauma operative management; ASSET, advanced surgical skills for exposure in trauma; BEST, basic endovascular skills for trauma.

\$187,000 to purchase all new simulators, with the cost reduced to \$21,500 using materials that were shared or developed in-house by the program.⁶³

American College of Surgeons Committee on Trauma Education Programs

Teaching and learning trauma patient management and trauma surgery present challenges because of the field's inherently unpredictable nature. The ACS CoT offers a series of simulation-based education programs to help address these challenges in training. The different programs are oriented toward learners with a range of experience, from community members to dedicated trauma fellows and practicing surgeons. The programs utilize a wide range of simulation modalities, including focused task trainers, high-fidelity manikins, animal models, and cadaver models. Dr. Mayur Narayan is a member of the Senior Educator Advisory Board (SEAB) of the ACS CoT which is responsible for the educational construct and delivery of all CoT courses. The team comprises 15 leading experts of surgical education across the globe. In addition, he is the Editor-in-Chief of the 4th edition of the ACS Trauma Evaluation & Management (TEAM) Course. The ACS CoT programs have been widely promulgated (Table 7). In a 2018 systematic review of open surgical skills training courses for trauma, 21 published courses were evaluated by Mackenzie and colleagues using Kirkpatrick's framework for evaluation of complex educational interventions.⁶⁴ Most courses had only Kirkpatrick level 1 (reactions) and level 2 (learning) evidence of benefit by participant self-evaluation. Only 3 courses—ATLS, Advanced Surgical Skills for Exposure in Trauma (ASSET), and Advanced Trauma Operative Management (ATOM)—all ACS CoT programs, had Kirkpatrick level 2 and level 3 (behaviors) evidence of benefit and evidence of skills retention over time. The CoT has active engagement of a large cohort of surgeons to continuously up-

date course materials and methods to ensure the programs remain current with updated trauma research and practice.

Advanced Trauma Life Support (ATLS)

The ATLS course was initially developed by Dr. James K. Styner, MD, after he experienced a personal tragedy in a traumatic accident. He and his family were involved in a plane crash in which his wife died at the scene.⁶⁵ He and his children were taken to a local rural hospital where they were evaluated by practitioners with little experience in trauma management. The experience drove him to develop a course to teach basic trauma evaluation and management. The basic ATLS “ABCDE” algorithm of Airway, Breathing, Circulation, Disability, and Exposure was developed by expert consensus and the course focused on identifying and treating immediately life-threatening injuries first. The course was presented to leaders of the ACS CoT in 1979. The course was approved by the ACS Board of Regents in 1980 and disseminated nationally, quickly becoming the standard in trauma evaluation and management in the United States and Canada. Eventually, the program was promulgated internationally and is now translated into 10 languages and delivered in 86 total countries.

The course content is updated roughly every 4 years with the 10th edition launching in 2017–2018 including a mobile-learning platform. Participants are provided the course manual and directed to review the content in advance of the course. During the course, trained faculty review highlights of the course content, lead the participants through clinical scenarios, and provide instruction in hands-on simulation of relevant procedures. Learners are assessed via a written examination and simulated trauma patient encounters on mannequins. Since its inception, ATLS has been the paradigm of trauma education and even served as an example for development of training courses in other domains, like the FLS curriculum.

Advanced Trauma Operative Management (ATOM)

The ATOM course was developed by Lenworth Jacobs, MD, MPH, in 1998. He developed the course content by recreating actual cases from the operating room to understand which problems could be represented by a porcine model.⁶⁵ The first course was offered in 2001. The course uses a porcine model to simulate common penetrating injuries to the chest and abdomen. The course is intended for senior surgical trainees and military or general surgeons in practice who may not frequently encounter these penetrating injuries. The course includes six 30-minute lectures and a 3-hour hands-on session during which the course participants manage 12 different injuries with one-to-one supervision by a course instructor. Dr. Jacobs and colleagues published results from the first 50 ATOM course students, which included expert traumatologists, general surgeons, trauma fellows, and fourth- and fifth-year general surgery residents.⁶⁶ All groups had improvement in knowledge and self-efficacy after the course. A survey of ATOM course participants in 2008 included 444 participants from 36 states and 17 countries.⁶⁷ Comments were largely positive, with approximately 80% to 90% of respondents indicating the course helped them identify and address injuries faster and more effectively. Ninety-nine percent of respondents felt the ATOM course is worthwhile. The ATOM course has also been described as an avenue for trauma team training. Perkins and colleagues describe development of a complementary course to ATOM for trauma OR staff.⁶⁸ Participating OR staff included nurses and surgical technicians. Participants had a didactic session in the morning addressing fundamental concepts of trauma surgery and OR preparation. They also had a chance to practice a trauma setup for the OR and assemble specialized equipment with an experienced trauma surgical technician. In the afternoon, the participants worked with surgeon participants in a simultaneous ATOM course as they managed the penetrating injury scenarios in the porcine model. Course participants showed a significant improvement in knowledge after the course and 90.5% reported they were “highly satisfied” with the course.

Advanced Surgical Skills for Exposure in Trauma (ASSET)

The ASSET course was developed by the CoT Surgical Skills Committee under the leadership of Drs. Luchette and Bowyer. The committee was established in 2005 specifically to develop

a standardized, skills-based course focused on exposure of anatomic structures whose injuries may be immediately threatening to life and limb. The committee developed their list of life and limb-threatening injuries for inclusion in the course using a modified Delphi process. They developed a course manual, standardized teaching slides, case-based clinical scenarios, and brief video demonstrations of the procedures to be performed by participants. The manual reviews key surgical exposures in the neck, chest, abdomen and pelvis, and upper and lower extremities. The 1-day course involves hands-on performance of the exposures on fresh human cadavers with close faculty (1:4 ratio) oversight and guidance.

The pilot course was conducted in March 2008 at the Uniformed Services University followed by a second course in October 2008 at the Stritch School of Medicine of Loyola University of Chicago. Kuhls and colleagues reported on the development of the course and outcomes of the first 4 beta courses.⁶⁹ Participants included senior surgical trainees, PGY4 and above. Participants had statistically significant increases in self-assessed confidence in operative skills in all anatomic regions included in the course and across all levels of training. In the overall course evaluation, participants' average rating across all questions was 4.73 out of 5, with "I gained new knowledge" at 4.80, "I learned new techniques" at 4.72, "I would recommend this course to colleagues" at 4.91, and "The amount of time allotted was appropriate" at 4.32. Beginning in March 2010, the ACS CoT has offered the course publicly for continuing medical education (CME) credit. Courses are now offered monthly across multiple states and countries. Bowyer and colleagues subsequently reported on the first 25 ASSET courses conducted after its final standardization in September 2010 through February 2012.⁷⁰ The students in those courses included 91 surgical trainees and 123 practicing surgeons. As with the beta courses, students reported a statistically significant improvement in comfort level or efficacy for all 25 ASSET skills that were reported from the pre and postsurveys. On aggregate course evaluation via 5-point Likert scale, participants responded with 4.83 out of 5 for "I gained new knowledge," 4.83 out of 5 for "I learned new techniques," and 4.92 out of 5 for "I would recommend this course to colleagues." An ASSET+ course has been developed specifically for military surgeons. The ASSET+ course includes the standard ASSET content, plus neurosurgery, obstetrics and gynecology, ophthalmology, and orthopedics. The course is expanded to 2 days and has a 1:1 trainee to instructor ratio.

Mackenzie and colleagues with the Retention and Assessment of Surgical Performance (RASP) Group of Investigators evaluated surgical trainees before, just after, and months after taking the ASSET course.⁷¹ Overall trainee performance improved when evaluated within 4 weeks after completion of the ASSET course. The improvement persisted on follow-up evaluation 12 to 18 months after ASSET training. Not surprisingly, interval experience during the follow-up period also affected trainee performance at the 12 to 18 month evaluation. The RASP group also evaluated a cohort of practicing surgeons, who were not dedicated traumatologists, at 30 months (on average) from the completion of an ASSET course.⁷² Overall, the practicing surgeons group had lower performance scores compared to residents who had completed the ASSET course more recently (an average of 14 months) and experts in trauma surgery. Across all 3 groups, poor performance correlated with low interval experience in performing the procedures. The study authors note that, in the absence of opportunities to perform the procedures in the clinical setting, skills training refreshers may be needed to prevent significant skills degradation.

Basic Endovascular Skills for Trauma (BEST)

The Basic Endovascular Skills for Trauma (BEST) course was initially developed by Dr. Megan Brenner and colleagues at the R. Adams Cowley Shock Trauma Center in Baltimore, Maryland. The first course was delivered to faculty in the Division of Trauma and Critical Care at Shock Trauma in 2013.⁷³ The 13 participants were board eligible or board certified general surgeons or emergency medicine physicians who took calls in the trauma resuscitation unit. Participants were required to be novice interventionalists, with no prior endovascular training, no participation in prior similar training courses, and no previous experience performing resuscitative endovascular balloon occlusion of the aorta (REBOA) in the clinical setting. Participants received didactic instruction on indications, steps, and pitfalls of REBOA along with percutaneous and open cannulation of the femoral artery. The participants then performed the procedure with in-

structors in a simulated setting. The initial course utilized a VR simulator, the Vascular Intervention System Training Simulator-C (Mentice VIST-C). The simulator allowed for virtual fluoroscopy to allow participants to visualize and better understand the procedure when performing REBOA with the instructor. Participants were then assessed on 3 attempts at placing the REBOA in zone I and 3 attempts at placing the REBOA in zone III using only virtual digital x-ray imaging, to reflect the resources of the actual clinical setting. Participants showed a significant improvement in knowledge from the pretest to the post-test and had a significant decrease in procedural task time from their first to their sixth attempts. All participants agreed that the course was beneficial, that it prepared them to perform the procedure on their next call, and that they would recommend the course to other acute care surgeons. In 2014, the course was opened to outside participants with wide interest resulting in 1 course per month for many years. In 2016, the BEST course joined the ACS CoT and Surgical Skills Committee. The BEST course has transitioned from VR to perfused cadaver models and includes training in endovascular techniques including REBOA as well as ultrasound-guided femoral artery access, percutaneous and open cannulation of the common femoral artery, and common femoral artery repair. As of 2021, BEST had been delivered to over 800 course participants in 25 states and Canadian provinces and 25 total countries.

Stop the bleed

Stop the bleed (STB) developed from a White House interagency workgroup in 2015 with a goal of increasing national resilience by preparing the public to stop life threatening bleeding. The phrase and logo are owned by the Department of Defense, which licenses them to the ACS, where the program is administered by the CoT. Unlike most CoT educational programs, STB is primarily targeted to nonmedical personnel, with nearly 4 million people worldwide trained since its inception. The STB course uses didactic materials and supervised hands-on practice on focused task trainers to teach 3 key techniques for hemorrhage control: manual pressure, wound packing, and tourniquet application. Within the medical community, the curriculum is increasingly being taught at the medical school level. Medical students have proven to be receptive to the curriculum, feeling empowered by the training to personally intervene on acute bleeding if necessary⁷⁴ and effectively serving as course instructors after their training.⁷⁵ Efforts are ongoing to optimize content delivery, with innovations including development of a perfused synthetic cadaver model to simulate active bleeding during the practical portion of the course.⁷⁶ At the ACS CoT annual meeting in 2018, Dr. Narayan raised the question on whether the 3 lifesaving skills of STB could be taught remotely to increase the dissemination. This suggestion was met with significant pushback as other trauma leaders insisted that a skills instructor would need to be present to teach the course effectively. During the pandemic era, Drs. Mayur Narayan and Ziad Sifri (Rutgers, New Brunswick & Rutgers Newark) studied video-based⁷⁷ and virtual live instruction⁷⁸ of STB, respectively, and were both shown to teach course content effectively. Finally, although developed as an initiative of the United States government, the STB course has been successfully implemented internationally, including in low- and middle-income (LMICs) countries in Asia, Africa, and South America.^{79,80} In 2017, Dr. Mayur Narayan helped bring the first STB course to India and has now helped disseminate the training to more than 20,000 police officers. This training was conducted with the assistance of SaveLIFE Foundation, a nongovernmental organization (NGO) that addresses road safety and trauma care in India and is the winner of the 2024 Skoll Awards for Social Innovation.

Virtual reality for surgical simulation

Pilots were regularly training in flight simulators. Many wondered if this same technology could help train laparoscopic surgeons in the new era of minimally invasive surgery. Washington University Institute for Minimally Invasive Surgery (WUMIS) drafted a grant to the Defense Advanced Research Projects Agency (DARPA) in 1994 and Dr. Dan Jones approached McDonnell Douglas aerospace company. The response was definitive, that surgery was much more complex than flight. There was not enough computer power or know-how to develop a simulator for surgery.

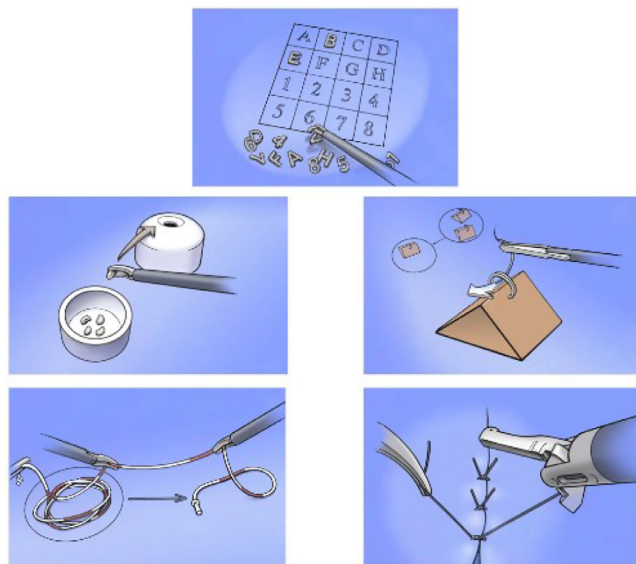


Fig. 11. Tasks: (A) checkerboard, (B) bean drop, (C) rope walk, (D) triangle transfer, (E) endo stitch.

Nevertheless, companies like Ethicon-Endo Surgery tried to develop the Preceptor, a laparoscopic cholecystectomy (LC) simulator. As a general surgery resident and consultant, Dr. Jones was convinced of the future of simulation. Eventually, the bulk of the simulator and cost became prohibitive to maintain. Surgery identified the potential for VR simulators but had to await computer software advancements.

In the late 1990s, laparoscopic video equipment (laparoscope, camera, light source) was rolled into the skills lab for learners to practice within a box that simulated the abdominal cavity.⁴ Tasks were developed from simple to more complex, with the ultimate goal to learn laparoscopic suturing (Fig 11).

At UT Southwestern, we used the Jones video trainers by Storz Endoscopy to develop skills.⁴ In a randomized study, we had PGY 2 residents perform a LC. One half of the trainees practiced 30 minutes per day for 2 weeks in the simulation center.³ The other half did their usual patient care activities. At the end of the month, everyone was observed by a blinded observer while they performed a LC. Trainees who practiced tasks improved, but they also performed measurably better in the OR. This was the first time we showed that practicing skills translated into improved OR performance.

The MIST VR was a commercial product that allowed trainees to use a laparoscopic instrument to do VR tasks.^{5,81} The original VR tasks would ask the learner to do tasks moving virtual objects, circles, squares, and cylinders (Fig 12). The tasks required eye hand coordination but did not resemble operative tasks at all. We performed a similar study whereby PGY 2 residents were observed doing a LC. One half of trainees were randomized to practice skills for 2 weeks. We again had a blinded observer score, and all residents performed a LC. Like before, those who practiced got better on tasks. They also got better in the OR, slightly more than what was seen training using video trainers despite trainees preferring the video trainers to MIST VR. We believe this may be that the VR system provided feedback on accuracy and precision or movements of right and left hand in contrast to video trainers that measured task completion times. In the real world, precision is likely more important than speed.

Over time, the VR platform allowed commercialization as more realistic tasks and operations were developed. In addition to LC, more complex operations like sleeve gastrectomy and gastric bypass (Fig 13) were being taught on virtual simulators.

Virtual Tasks

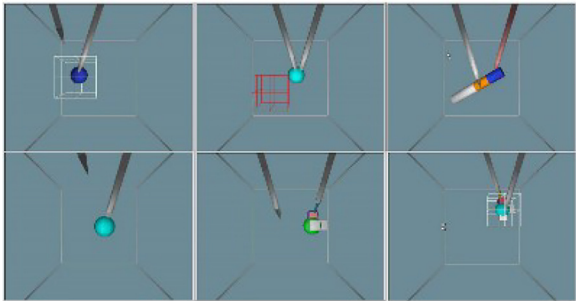


Fig. 12. Minimally invasive surgical trainer virtual reality (MIST-VR) tasks.

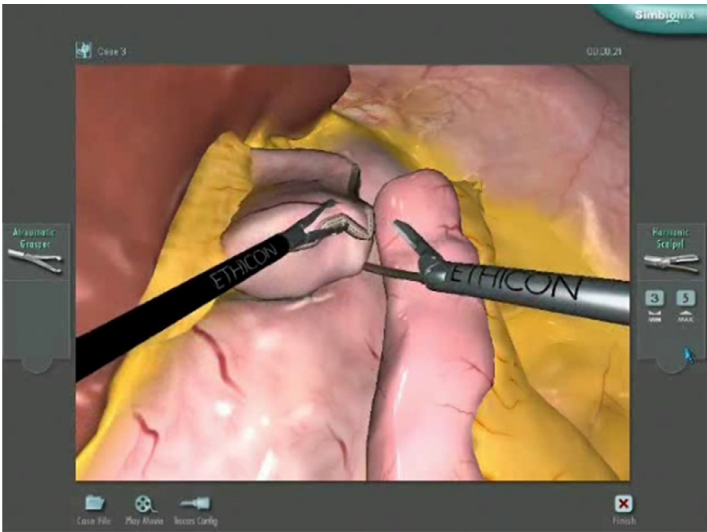


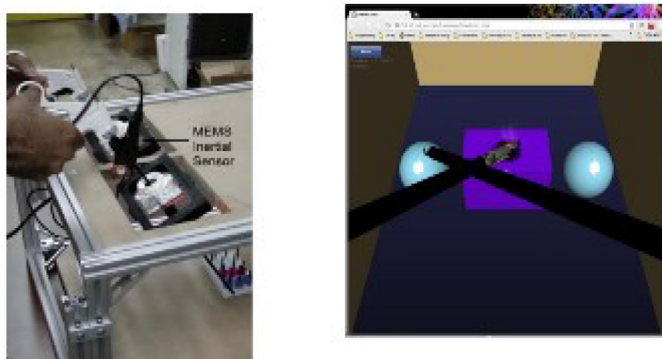
Fig. 13. Virtual reality Roux-en-Y gastric bypass.

SAGES embraced the GI Mentor as the platform for the FES certificate and popularized the sale of the GI Mentor as FES Testing Sites purchased simulators for testing. Once the ABS mandated FES certification prior to sitting for the board examination in surgery, many residency programs purchased the GI Mentor to allow residents to practice for the examination.

Our research group focused our efforts on developing simulators for surgery using VR. In collaboration with Dr. Suvrnu De (Professor at RPI and now Dean at Florida State University), we first looked to develop VR versions of the FLS exam. We developed a VR endoloop, peg transfer, circle cutting, and suturing tasks.⁸²⁻⁸⁹ VR lacked tactile realistic tactile feedback, but it allowed real time feedback on task performance. The actual FLS examination required sending looped foam, cut gauze, and sutured penrose to be sent out to be scored for quality and accuracy. A second advantage of VR, is there were not a lot of expensive consumable product required such as endoloops, sutures, and penrose. VR also proved useful for teaching key operative steps such as passage of the laparoscopic band⁹⁰ and graded complexity such as a difficult airway (Fig 14)⁹¹⁻⁹³ and confusing concepts such as spread of radiofrequency energy (Fig 15)^{48-50,94} and response to a VR OR fire.^{11,95,96}



Fig. 14. Virtual reality difficult airway.



Sankaranarayanan G, Bachun L, Miller A, Wakily H, Jones SB, Schwaizberg S, Jones DB, De S, Olasky J. Face validation of the virtual electrosurgery trainer (VEST). *Surg Endosc*, 30(2):730-8, 2016

Fig. 15. Fundamental Use of Surgical Energy (FUSE) virtual reality simulator.

Some of the most popular commercial VR simulations have been developed for learning robotic surgery. This teaches both tasks and operations such as robotic inguinal and ventral hernia operations (Fig 16). The Fundamentals of Robot Surgery uses simulation as the framework to teach robotic surgery. Gamification has been used to engage resident learners.⁹⁷

The holy grail of VR was the idea of creating a Star Trek Holodeck for Surgery. The head mounted viewing system proved very immersive and as computer power skyrocketed engineers were able to tap gaming software to develop amazing VR simulations for the first time.^{10,98} The surgeon could walk into a VR OR as an avatar and perform gradually more complex operations and scenarios (Fig 17). As the VR gaming industry advanced programming, around 2015 we had computer power to begin developing more immersive VR learning. Over the last 20 years we had been doing team training using manikins; now only our imagination limited crisis scenarios.

Our group developed a virtual mock OR.¹⁰ We decided to study how surgeons, anesthesiologists, and OR nurses responded to an OR fire. Although most providers believed they had sufficient experience to deal with an OR fire from the Association of Perioperative Registered Nurses (AORN) and FUSE courses, few had ever experienced an actual OR fire. There are estimated to be approximately 500 OR fires annually in the United States. In our study, a provider



Fig. 18. Multiple avatars communicating and acting together.

wears an Oculus Headset and is immersed in a VR OR. The drapes catch fire, and the participant has choices as to how to manage the crisis while listening to sirens and vision obscured with virtual smoke. When we tested 185 surgeons, anesthesiologists, and nurses, less than 5% passed and turned off the oxygen feeding the fire.^{11,95} With repetition, almost all improved performance. This also highlights how VR simulation can do scenarios that would be costly, timely, and dangerous as an actual simulation versus a virtual simulation.

We have found that in addition to immersion, VR allows a medium for teaching complex concepts. For example, we wanted to teach surgical trainees about the principles of energy, specifically for FUSE.^{48-50,94} In 1 study, we wanted to show the path of radiofrequency energy from the Bovie tip to the dispersive electrode.⁹⁴ In another study, we used VR to demonstrate the principle of capacitance coupling, charge, insufflation failure, and sparks, depending on how laparoscopic instruments are used.⁴⁸

Our current VR research seeks to explore how OR teams work together to solve complex problems during a crisis. Instead of 1 Oculus and 1 avatar, we are now seeking multiple aviators (Fig 18). We can study decision-making of the surgeon, anesthesiologist, first assistant, and nurse (Fig 19). Who speaks up? Is a checklist used? How does hierarchy, gender, or racism hinder communication? How can we strengthen team communication? VR can create a safe space to learn and develop technical and nontechnical skills. The virtual platform facilitates training, innovation, and research.

Artificial intelligence and its integration in surgery

AI possesses capabilities that mirror human cognition and judgment. As global volumes of treatable surgical conditions, such as cancer, injuries, and noncommunicable diseases⁹⁹ continue to rise, the integration of AI into surgical workflows presents a promising solution to overcome the limitations of infrastructure, training, and human capacity.^{100,101} AI algorithms, utilizing population-level data, already guide surgical risk assessments and enhance the capabilities of robotic surgery, with numerous studies demonstrating that their performance can exceed or complement that of human experts in specific medical use cases.^{102,103}

Despite its potential, AI applications in surgery have often been narrowly focused on specific diseases or standalone procedures. However, by leveraging the diverse strengths of AI technologies—such as machine learning (ML), natural language processing (NLP), computer vision (CV), and artificial neural networks (ANN)—a broader and more integrated approach is



Fig. 19. Surgeon interacting with multiple avatars in a virtual operating room.

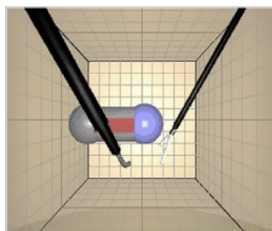
emerging. These technologies, when applied with insight and precision, promise to transform surgical education and practice. The importance of surgical education cannot be overstated, as it is fundamental to preparing skilled surgeons capable of meeting the growing global demand for surgical care. As AI tools continue to evolve, they herald a paradigm shift in surgical education, offering new ways to enhance learning, improve outcomes, and expand the capabilities of surgeons. This transformation holds the potential not only to improve surgical proficiency but also to ensure that high-quality surgical care is accessible to all who need it.

Several FDA-approved AI tools, underpinned by ML, are already making significant strides across various stages of patient care.¹⁰⁴ For instance, ML algorithms have demonstrated greater accuracy than traditional clinical guidelines in predicting lung cancer staging.¹⁰⁵ Additionally, they have been effectively applied for surgical imaging analysis, decision support, and operative planning. Other applications include ML-integrated triage tools that identify signs of inadequate organ perfusion and automatically alert surgical teams to intervene.¹⁰⁶ At a patient-centric level, ML models can accurately gauge the efficacy of interventions, such as predicting responses to fluid administration in kidney injury scenarios.¹⁰⁷ Although classical scoring systems still assist with risk assessment and surgical planning, they relate patient factors linearly and additively. In contrast, ML models can capture intricate multidimensional relationships and interactions between variables.¹⁰⁸ This depth of analysis paves the way for enhanced surgical decision-making, workflow efficiency, and predictive accuracy, all culminating in improved patient outcomes.

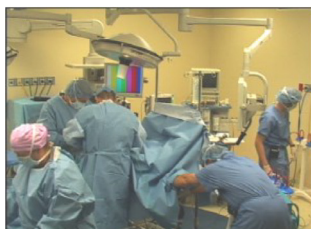
CV enables systems to interpret and understand visual data, thus unlocking an array of possibilities from enhancing training and education, operative complexity analysis, operative assessment and feedback, and finally intraoperative decision support.¹⁰⁹ By processing high-resolution surgical video feeds, CV algorithms can identify anatomical structures, surgical tools, and steps in a procedure.¹¹⁰ Moreover, CV algorithms can provide automated assessment and feedback on surgical videos to augment coaching and skill development.¹¹¹ The Virtual Operative Assistant (VOA), for instance, utilizes CV for comprehensive simulation-based learning, breaking down intricate procedures.¹¹² Additionally, CV shows promise for improving surgical decision-making through surgical phase recognition; algorithms can process videos to detect expected versus un-



A



B



C



D

Fig. 20. (A) Video trainer. (B) Virtual reality tasks. (C) Mock operating room team training. (D) Virtual operating room.

expected steps and alert surgeons.¹¹³ Multiple groups have developed CV models to segment anatomy, recognize instruments, and classify surgeon skill level with better than 90% accuracy by analyzing video data.¹¹⁴ Furthermore, "surgical fingerprinting" identifies deviations from standard procedures, potentially highlighting complication-prone areas.¹¹⁵

Much like navigating the dense molecular signaling of neurons, powerful ANN algorithms can trace hidden relationships across vast datasets of surgical cases and patient factors.¹¹⁶ ANNs

Table 8
Benefits and challenges of artificial intelligence.

Benefits	Challenges
Enhanced learning experience - Personalized education - Adaptive learning	Technological limitations - Integration with existing system - Data quality and availability
Improved skill assessment - Objective evaluation - Standardized metrics	Ethical and legal concerns - Privacy and security - Bias and fairness
Simulation and virtual reality - Realistic training environments Rare case simulation	Resistance to change - Adoption by educators and trainees Cultural and institutional barriers
Data driven insights - Performance analytics - Predictive analytics	Cost and resource allocation - High initial investment - Ongoing maintenance and upgrades
Enhanced collaboration and communication	Ensuring reliability and accuracy

have revolutionized in-hospital mortality prediction after hepatic malignancy surgeries, achieving higher accuracy than traditional logistic regression techniques.¹¹⁷ This breakthrough significantly impacts patient outcomes in regions where hepatocellular carcinoma is highly prevalent. In cholecystectomy, ANNs function as expert systems, streamlining workflows and improving concurrent diagnoses.¹¹⁸ In addition, ANNs show tremendous promise in medical imaging analysis, with applications ranging from tumor detection in mammography to tissue classification in magnetic resonance imaging.¹¹⁹ Together, these diverse use cases showcase the substantial capabilities of ANNs to enhance risk analytics and resource optimization, optimize treatment strategies, and reduce diagnostic uncertainty through their unparalleled ability to model multi-dimensional datasets.

Lastly, *NLP* enables nuanced computational understanding and generation of human language. It is already demonstrating immense potential to transform surgical care through enhanced clinical documentation by automating the extraction of diagnoses and procedure codes from electronic health records, ensuring both time efficiency and accuracy.¹²⁰ Additionally, *NLP* tools exhibit high performance in analyzing clinical narratives like radiology reports to identify key findings.¹²⁰ *NLP* also shows promise for optimizing evidence-based decision-making by efficiently extracting insights from literature to provide real-time guidance. Beyond documentation, *NLP* could even change surgeon-computer interaction through virtual scribes that listen and translate OR dialog.¹²¹ And for education, *NLP* systems can parse academic texts and surgical conversations to create comprehensive surgical educational resources.¹⁰⁹ Overall, these diverse applications showcase *NLP*'s capabilities to streamline workflows, advancing clinical documentation, enhancing decision support, democratizing research, and ultimately improving patient outcomes through comprehending nuances in language.

A harmonized blend of technology, complemented by the combined endeavors of surgeons, researchers, and tech specialists, set the stage for enhancing surgical training, including personalized learning, objective skill assessment, and realistic simulations while also presenting challenges such as technological limitations, ethical concerns, and resistance to change. [Table 8](#) summarizes the benefits and challenges of using AI in surgical education. Addressing these challenges through careful planning, investment, and continuous evaluation will be crucial to fully harness the potential of AI in transforming surgical education.

Collection and analysis of surgical data

The intraoperative period is a data-rich environment, providing an invaluable resource for surgical education and simulation.^{103,122} Continuous monitoring of physiological parameters

amid complex alterations to anatomy and physiology makes this time the core of surgical practice. As discussed previously, advances in intraoperative CV have enabled significant progress in the analysis of anatomy, including the assessment of tissue characteristics, dissection planes, and pathology identification.^{111,113} Additionally, the reliable identification of instruments and stages of operation, as well as the prediction of procedural next steps, are crucial for developing future autonomous systems and data-driven improvements in surgical techniques.

Events inside the OR have substantial impacts on recovery, postoperative complications, and oncological outcomes. Despite their pivotal importance, minimal data are currently recorded, analyzed, or collected in this setting.¹²³ Additionally, visual analytics tools and methodologies play a vital role in presenting complex data in an understandable format, facilitating better decision-making and comprehension among surgical trainees.¹²⁴ Thus, harnessing valuable data streams from the intraoperative period can significantly contribute to advances in surgical automation and underpin the utility of AI in the OR, especially for educational purposes.

In the context of surgical education and simulation, the integration of various data sources becomes even more critical.

Electronic health record. The EHR integrates patient histories, laboratory results, and imaging studies to support surgical planning and outcomes analysis. These data can be used to create realistic, patient-specific scenarios in simulation training.

Intraoperative video recordings. Intraoperative video footage can be used for performance review and training. These recordings can serve as a valuable resource for trainees to study surgical techniques, decision-making processes, and intraoperative management.

Intraoperative variables. Monitoring variables such as the length of the procedure, duration of anesthesia, and OR throughput can help in understanding the efficiency and effectiveness of surgical practices, which can then be incorporated into simulation scenarios to enhance training.

Intraoperative biomedical images. Using real-time imaging data can be used to educate trainees on recognizing anatomical structures, pathology, and surgical landmarks. These data can be incorporated into virtual and augmented reality simulations to provide an immersive learning experience.

Sensor data. Data from surgical instruments, wearables, and smart devices to monitor surgeon movements and patient vitals can be used to create detailed simulations that mimic real-life surgical conditions, allowing trainees to practice and refine their skills in a controlled environment.

Simulation data. Performance metrics from virtual reality and simulation-based training environments can be analyzed to identify areas for improvement, tailor training programs to individual needs, and track progress over time.

By leveraging these data sources, surgical education can be significantly enhanced, providing trainees with a comprehensive, data-driven learning experience. Just like others, surgical data acquisition should adhere to best practices for gathering high-quality data from various surgical and clinical sources. Effective data management strategies are essential for storing, organizing, and securing large datasets, ensuring they are accessible for analysis while maintaining patient confidentiality.

Using surgical data to benchmark performance against best practices and standards is essential for facilitating continuous improvement. Data-driven feedback can help surgeons understand their strengths and areas for development, leading to more effective training and professional growth.¹²⁵ Implementing learning health systems, where data are continuously analyzed to generate new insights, can improve surgical practices in real-time. It is important to avoid data misuse by preventing unnecessary competition that could lead to burnout and decreased quality of care. Ethical research and innovation should focus on using data to benefit the broader medical community and patients, ensuring that advancements derived from data analysis directly contribute to improved surgical outcomes.

Responsible data stewardship involves fostering a collaborative environment where data are shared among institutions and practitioners, promoting collective improvements in surgical practices rather than unhealthy competition. Transparency and accountability are critical; clear

guidelines for data use must ensure all stakeholders are aware of how data are collected, analyzed, and applied, with oversight from data governance committees to maintain ethical practices and compliance with standards.¹²⁶ Lastly, informed consent and privacy are critical when using patient data for educational purposes. Maintaining rigorous standards for obtaining patient consent and protecting their privacy throughout the data lifecycle ensures ethical data use.¹²⁷ The ultimate goal is to ensure that data utilization directly or indirectly benefits patients by improving surgical outcomes.

Artificial intelligence-powered surgical simulation and virtual reality training

In 1997, a study involved a blinded evaluator assessing PGY-2 trainees performing a LC by grading their performance.³ Given that the operations were video recorded, the researchers attempted retrospectively to grade performance using the recordings. However, they quickly discovered that locating the relevant video clips was exceedingly time-consuming.¹²⁸ Additionally, the absence of audio and external views hindered their ability to accurately determine the actions of the operating surgeon and the assistant, as well as any verbal cues provided. Consequently, they concluded that video capture without external views and audio recording was neither easier nor superior.

Today, the landscape has transformed significantly. Image quality and video capture quality has vastly improved, and advanced technologies such as AI now facilitate the selective capture of key parts of the operation. Theator Inc. is 1 example of a company operating in this domain.¹²⁹ By the time an LC is completed, AI has already selected the critical video segments, enabling the attending surgeon to review these clips with the trainee. Beyond postoperative feedback, AI is also being investigated for its potential to guide operative decision-making by overlaying colored indicators of safe and dangerous areas for dissection.¹³⁰

Many groups are attempting to store clips as part of the video operative note. Other groups have shared video clips on social media like Twitter and Facebook channels. The Robot Surgery Collaboration has more than 14,000 surgeons online to opine on uploaded video clips.¹³¹ Many believe that the ABS and hospital privileging board will 1 day require operative video-based assessments.¹³² In principle, AI could learn to select and grade key steps of specific operations.

As such, the integration of AI in VR and augmented reality (AR) is transforming surgical education by providing immersive training environments where realistic simulations of surgical procedures can be performed in both low and high resource settings.¹³³ These technologies allow for a heightened sense of realism through haptic feedback and tactile sensations, essential for practicing and refining surgical skills. Customizable scenarios with varying levels of difficulty enable tailored learning experiences, and performance tracking along with real-time feedback further enhances the educational value of these simulations.

AI-powered VR training platforms significantly improve operative performance and reduce operative times by providing proficiency-based training with immediate feedback and on-the-spot instructions. This method promotes deliberate practice and enables the rehearsal of procedures at varying difficulty levels, thereby improving learning outcomes. The incorporation of VR training into surgical curricula is being advocated, provided its cost-effectiveness is demonstrated.¹³⁴ Moreover, AR is increasingly being used to boost surgical visualization and guidance through technologies such as fluorescence-guided surgery and innovative imaging devices like contrast-enhanced ultrasound and photoacoustic imaging. These advancements offer surgeons enhanced visualization of intraoperative anatomy, crucial for precise and safe surgical interventions.¹³⁵

VR simulators, combined with ML, enable the quantification of multiple aspects of surgical performance in a safe, controlled environment. Supervised classifiers, which use prelabeled data to generate predictive models, and unsupervised algorithms, which identify patterns without prior labeling, are both utilized to assess and categorize surgical expertise. To standardize and

Table 9
The Machine Learning to Assess Surgical Expertise (MLASE) checklist contains 20 elements, separated into 4 sections. A point is awarded for every element completed in the article. The total score is calculated by adding the total number of elements checked.

Section	Element	Yes?
Study design (5 points)	1. Is relevant literature on the use of artificial intelligence in simulation provided?	
	2. Is the sample size clearly stated (including number of groups and number of participants in each group)?	
	3. Is a definition of each group of expertise provided?	
	4. Is the simulator described?	
	5. Are the surgical tasks to be performed outlined?	
Data structure (6 points)	6. Is raw data acquisition described?	
	7. Is feature extraction mentioned?	
	8. Is an effort made to normalize the data?	
	9. Is feature selection mentioned?	
	10. Is the count of features used by the algorithm clearly stated?	
Supervised machine learning (5 points)	11. Are the final selected features clearly described?	
	12. Is the type of the classifier used mentioned and justified (either by comparing multiple classifiers or citing relevant literature)?	
	13. Is the mechanism of the classifier explained or is a relevant source provided?	
	14. Is an effort made to clearly describe the methods used to train and test the algorithm?	
	15. Is the accuracy of the classifier mentioned?	
Discussion quality (4 points)	16. Is the sensitivity and specificity mentioned?	
	17. Are efforts made to explain the educational rationale of the features used by the algorithm?	
	18. Is the educational application of classifiers in the context of surgical simulation stated, specifically its use as a summative or formative assessment tool?	
	19. Are methodological limitations discussed, including those pertaining to any above points?	
	20. Are the future directions discussed?	
Total Score = ____/20		

improve the reporting of ML-assisted surgical education studies, a framework known as the Machine Learning to Assess Surgical Expertise (MLASE) checklist has been developed. This checklist (Table 9) ensures the quality and comprehensiveness of research in this field by providing clear guidelines on study design, data structure, the use of supervised ML, and discussion quality.¹³⁶

In summary, AI-powered virtual reality and augmented reality simulators represent a transformative advancement in surgical education. By providing immersive, realistic, and customizable training environments, coupled with the analytical power of machine learning, these technologies hold the promise of significantly enhancing surgical training and, ultimately, patient care. However, addressing the challenges of data privacy, standardization, curriculum integration, and ethical considerations will be crucial for their successful implementation and sustained impact.¹³⁷

Dr Abby Susk summarized, “Responsible AI and technology governance is something we should all care about. I’m concerned about the false dichotomy between developing and deploying AI responsibly and creating products that are competitive on the market or in the international arena. Responsible AI work ensures that end-users trust that systems will operate as promised, have an understanding of how they work, and know that using the system will not introduce legal consequences or moral injury. Achieving those outcomes facilitates adoption of the AI system by end-users, and is thus essential. People will not leverage a technology that they do not trust or understand.”¹³⁸

Artificial intelligence-assisted surgical planning and 3D modeling for procedural efficiency

Early efforts to collect data in the operating room include initiatives like the OR Black Box¹³⁹ designed to provide a dependable system for auditing and monitoring intraoperative events and practice variations. One particularly innovative advancement aimed at enhancing surgical teamwork is the preliminary development of an AI coach. This AI coach is designed to infer the alignment of mental models within a surgical team.¹⁴⁰

Multimodal digital inputs, such as physiological data (eg, skin conductance and heart rate variability) to identify operative stress, continuous pharmacological and vital signs monitoring from anesthesia, and inputs from nursing team staffing and equipment availability, can be quantified and integrated into a digital pathway for automation and optimization in surgical education. The use of generative AI models incorporating both patient and environmental data in the operating room enhances nontechnical skills crucial for surgical trainees, including communication, situational awareness, and team functioning.¹⁴¹ Tracking factors like operative fatigue, miscommunication between anesthesiologists and surgeons, staffing changeovers, and equipment shortages can reduce intraoperative errors. A digitized surgical platform thus holds the potential to facilitate an AI-enhanced future in surgical education.

Case study 1: Osso VR and surgical education¹⁴²

Methods. This prospective, blinded study involved 38 fourth-year medical students with no prior experience in tibia intramedullary nail (IMN) surgery. Participants were randomized into 2 groups: 1 trained using a traditional technique guide and the other using the OSSO VR platform. The VR module developed by OSSO VR Inc. provided an immersive experience, with audio, visual, tactile, and proprioceptive feedback, simulating a virtual OR. The participants' performance was evaluated based on the time to completion of the mock surgery and the number of corrections needed. A blinded observer used the Global Assessment 5-point Rating Scale and a Procedure-Specific Checklist to objectively grade the participants.

Evaluation of skills. Participants in the VR group completed the mock surgery in significantly less time (9.6 minutes vs. 12.2 minutes, $P=0.034$) and required fewer corrections (2.2 vs. 2.5; $P=0.05$) compared to the traditional group. The VR training also improved the participants' proficiency in the steps of the tibia IMN procedure, indicating a better overall understanding and execution of the surgical technique.

Results. The study found trends in favor of the VR group across most categories. Although some differences in metrics did not reach statistical significance, the VR group consistently outperformed the traditional group in key areas such as time to completion and the number of procedural corrections. The improved performance in VR training suggests that it may offer a more effective method for surgical education, particularly in learning complex procedures like tibia IMN surgery.

Discussion. The results indicate that VR training can enhance surgical proficiency by providing an immersive, interactive learning environment that replicates the OR experience. This method allows trainees to practice and refine their skills in a risk-free setting, potentially reducing the learning curve and improving patient outcomes. The study highlights the potential of VR to augment traditional surgical training, offering a valuable tool for medical educators. However, the study also acknowledges limitations, such as the small sample size and the use of medical students rather than surgical residents. Future research should explore the long-term retention of skills acquired through VR training and its impact on actual surgical performance and patient outcomes.

Conclusion. VR, exemplified by OSSO VR, presents a promising training modality for orthopedic trauma surgery. The study demonstrates that VR training can enhance surgical skills, reduce procedure times, and decrease the need for corrections, suggesting its potential to improve surgical education and patient care. Further research with larger cohorts and long-term follow-up is warranted to fully establish the efficacy of VR in surgical training.

Artificial intelligence for skill acquisition and retention

In this section we explore dynamic scenario generation, intelligent tutoring systems, adaptive learning, and personalized surgical education, highlighting their roles in modernizing surgical training.

Dynamic scenario generation involves creating realistic and varied training scenarios that simulate the complexity and unpredictability of real-life surgical situations. High-fidelity simulations using VR and AR provide immersive environments that closely mimic actual surgical procedures. These scenarios expose trainees to a wide range of surgical challenges, including rare and complex cases, thereby improving their decision-making and problem-solving skills. Immediate feedback from these simulations allows trainees to learn from their mistakes and refine their techniques in a risk-free environment. Carla Pugh's work involves integrating VR into surgical training to create immersive, interactive learning environments. Her research utilizes haptic feedback systems to simulate the tactile sensations of real surgical procedures, allowing trainees to develop a sense of touch and pressure that is crucial for surgical proficiency.¹⁴³ Pugh's VR simulations using sensor technology are designed to adapt dynamically to the trainee's actions, providing a realistic and engaging training experience.^{144,145} For example, in a VR simulation of laparoscopic surgery, the system can simulate variations in tissue resistance and bleeding, requiring the trainee to adapt their technique accordingly.¹⁴⁵ These simulators offer realistic training scenarios and track performance over time, helping trainees maintain their surgical skills.¹⁴⁶

Intelligent tutoring systems (ITS) are automated teaching platforms that offer individualized performance feedback using advanced technologies such as machine learning and AI.¹⁴⁷ These systems deconstruct psychomotor skills into teachable metrics, providing specific, actionable feedback on various aspects of surgical performance. By leveraging large datasets, ITS can differentiate levels of surgical expertise and track progress over time, helping trainees understand their strengths and areas for improvement. This continuous, iterative practice with decreasing reliance on human supervision enhances trainees' skills through repeated exposure to complex tasks.

Case study 2: intelligent tutoring systems¹⁴⁸

One of the most groundbreaking implementations of ITS in surgical education is the Virtual Operative Assistant (VOA), developed for neurosurgical training. The VOA represents a significant advancement in how neurosurgical skills are taught, providing a detailed, data-driven approach to skill acquisition and assessment.

Development and implementation. The VOA integrates advanced ML algorithms and VR technology to create an immersive training environment. Developed at the Neurosurgical Simulation and Artificial Intelligence Learning Centre at McGill University, the system was designed to address the complex needs of neurosurgical education. Neurosurgery requires precise, fine motor skills and deep anatomical knowledge, making it an ideal setting for high-fidelity simulation and intelligent tutoring.

Functionality and features. The VOA system deconstructs complex neurosurgical procedures into discrete, teachable components. It uses VR to simulate the operative environment, allowing trainees to interact with hyper-realistic anatomical models and surgical tools. The ML algorithms analyze the trainee's movements and decisions in real time, providing immediate feedback on performance.

Key features of the VOA include:

- (1) *Hyper-realistic simulations.* The VR environment mimics the tactile and visual feedback of real surgical procedures, enhancing the realism of the training experience.
- (2) *Detailed performance metrics.* The system tracks a wide range of metrics, including hand movements, tool usage, and procedural accuracy. These metrics are used to provide detailed, personalized feedback to the trainee.

- (3) *Adaptive learning pathways.* Based on the performance data, the VOA adapts the training modules to focus on areas where the trainee needs improvement. This ensures a tailored learning experience that evolves with the trainee's skill level.
- (4) *Automated feedback.* The VOA provides immediate, data-driven feedback on technical performance, allowing trainees to make adjustments and *improve* their skills continuously.

Educational impact. The introduction of the VOA has had a profound impact on neurosurgical training. Studies conducted at McGill University have shown that trainees using the VOA demonstrate significant improvements in technical skills compared to those using traditional training methods. The system's ability to provide objective, standardized assessments has also been praised for reducing variability in skill evaluation.

Specific educational benefits include:

- (1) *Enhanced skill acquisition.* Trainees are able to practice complex procedures repeatedly in a risk-free environment, leading to faster and more effective skill acquisition.
- (2) *Objective assessments.* The data-driven feedback provided by the VOA offers an objective measure of technical proficiency, *reducing* reliance on subjective evaluations by human instructors.
- (3) *Increased confidence.* By allowing trainees to practice and perfect their skills before performing actual operations, the VOA helps build confidence and reduces anxiety associated with high-stakes procedures.

Integration into training programs. The successful integration of the VOA into neurosurgical training programs requires dedicated learning centers equipped with the necessary VR and computing infrastructure. At McGill University, a dedicated simulation lab has been established, providing trainees with access to the VOA and other advanced simulation tools. This laboratory is designed to facilitate iterative practice and continuous skill development, with minimal need for direct supervision by faculty.

Challenges and considerations. Although the VOA represents a significant advancement in surgical education, its implementation is not without challenges. The development and maintenance of such sophisticated systems require substantial financial investment. Ensuring the system's integration with existing training curricula and obtaining buy-in from educators and trainees are also critical steps. Additionally, addressing ethical and privacy concerns related to data collection and usage is paramount.

The VOA exemplifies the transformative potential of intelligent tutoring systems in neurosurgical training. By providing realistic simulations, detailed performance metrics, and adaptive learning pathways, the VOA enhances the educational experience and prepares trainees for the complexities of real-world surgery. As technology continues to advance, tools like the VOA will play an increasingly vital role in shaping the future of surgical education, ensuring that the next generation of surgeons is well-equipped to deliver high-quality patient care.

Adaptive learning technologies customize educational content based on the learner's performance, ensuring that training is both efficient and effective. AI algorithms tailor training modules to address individual needs, focusing on areas where the trainee requires more practice. The system dynamically adjusts the difficulty and focus of training exercises in real time, promoting optimal learning outcomes. This scalable solution can be applied across various levels of surgical education, from novice to expert, ensuring that each trainee receives the appropriate level of challenge and support. Hung and colleagues conducted longitudinal studies to track surgical trainees' performance over time. By collecting data from multiple procedures, they analyzed trends and patterns to understand skill development and retention.^{149,150,151} Their AI-powered predictive models forecasted a trainee's future performance based on past data. These models helped identify when a trainee might need additional training to prevent skill degradation, ensuring continuous improvement and maintenance of surgical skills.

Personalized surgical education aims to cater to the unique learning needs of each trainee, enhancing his/her overall educational experience. This involves developing customized training plans that align with the trainee's goals, strengths, and areas for improvement. Access to men-

tors and experienced surgeons who provide guidance, support, and feedback is crucial throughout the training process. Personalized education focuses not only on technical skills but also on nontechnical skills such as communication, teamwork, and leadership, ensuring holistic development.

To support this personalized approach, software applications developed specifically for use on small, wireless devices, like smartphones and tablets, can provide cognitive skills training through simulation apps. Examples include iLappSurgery, Touch Surgery, and Think Like A Surgeon.¹⁵² These apps offer interactive and accessible platforms for trainees to practice and refine their skills anytime and anywhere, further enhancing the flexibility and effectiveness of their surgical education.

Artificial intelligence in intraoperative guidance and assistance

Identification of anatomical structures and object recognition

Multimodal AI interrogation of the surgical field significantly aids in identifying relevant and aberrant anatomy, enhancing the surgeon's visual assessments, determining the necessity for biopsies, and evaluating malignancy risks.¹⁰³ Additionally, the benefits of AI in hypothetico-deductive surgical decision-making are well-documented.¹⁵³ Many groups across the world have successfully developed algorithms that identify anatomic surgical objects in the field and surgeon gestures using CV technologies.^{154,155,156} Some of the key contributions in the field are recognized below.

The AI-powered surgical training program at the New Jersey Institute of Technology, being developed by Associate Professor Usman Roshan and collaborators, exemplifies the utility of such advancements in surgical education. This program uses CV to analyze live video feeds of surgical exercises, offering immediate, personalized feedback. It aims to enhance the training process by reducing the need for constant supervision and manual feedback, with plans for integration into surgical curricula by 2025.¹⁵⁷

Key studies and applications

AI for predicting surgical events. A study by Ward and colleagues explored the use of AI to predict intraoperative events during cholecystectomy.¹¹⁵ The AI model analyzed surgical videos to identify patterns indicative of potential complications, showing promise in predicting adverse events, thereby allowing surgeons to take preemptive measures to mitigate risks.

Automated recognition of surgical steps. Hashimoto and colleagues developed an AI system capable of recognizing surgical steps in real-time during laparoscopic sleeve gastrectomy.¹⁵⁸ The model utilized DL algorithms to accurately classify each step of the procedure, enhancing intraoperative workflow and providing real-time feedback to surgeons.

AI for anatomical landmark identification. Tokuyasu and colleagues created an AI system using DL to identify anatomical landmarks during LC.¹⁵⁹ This system significantly improved the identification of critical structures, reducing the likelihood of accidental damage and improving surgical precision.

Assessment of the critical view of safety (CVS). Mascagni and colleagues introduced a computer vision platform that automatically assesses the CVS during LC.¹⁶⁰ The AI model evaluated the CVS, a crucial step to ensure the correct identification of the cystic duct and artery, thus preventing bile duct injuries. The system provided real-time feedback, helping surgeons achieve and verify the CVS more reliably.

Object detection tasks. DL models output bounding boxes and labels for individual objects within images.¹⁶¹ This technology is also useful in medical fields such as polyp detection during colonoscopy, where it has the potential to reduce human error. In surgery, AI-assisted surgical

instrument detection has been reported, where kinematic information about a surgeon can be utilized for workflow recognition and skill assessment.¹⁶² Intraoperative tracking of surgical instruments is a prerequisite for computer- and robotic-assisted interventions, and object detection approaches can effectively extract such data.

Overall, AI-driven systems are increasingly being integrated into operating rooms to assist surgeons in real time. These systems leverage CV and ML algorithms to analyze surgical videos, identify anatomical structures, and provide actionable insights during procedures.¹⁶³ The primary goals are to enhance situational awareness, support decision-making, and minimize errors. By utilizing these advanced technologies, surgical education and training programs can significantly improve, providing trainees with immediate feedback and fostering a deeper understanding of complex surgical procedures.

Case study 3: GoNoGoNet^{130,164}

Introduction. LC carries a risk of bile duct injuries, which can result in significant morbidity. Such injuries often occur due to misinterpretation of anatomical landmarks during surgery, exacerbated by inflammation or aberrant anatomy. AI, particularly deep learning models, offers the potential to augment the surgeon's mental model by providing real-time guidance to minimize these risks. The GoNoGoNet, is a CV algorithm designed to identify anatomical structures and delineate safe ("Go") and dangerous ("No-Go") zones during surgeries. This case study explores the development, implementation, and impact of GoNoGoNet in LC.

Development process. GoNoGoNet is a convolutional neural network (CNN) specifically trained to identify Go and No-Go zones during LC. The development process involved:

- (1) *Data collection.* The training dataset comprised 308 anonymized LC videos from 37 countries, involving 153 surgeons and 136 institutions. The videos included various conditions such as acute and chronic cholecystitis and required lysis of adhesions, providing a comprehensive dataset for training the AI model.
- (2) *Annotation process.* Expert surgeons annotated the videos to mark the Go and No-Go zones, liver, gallbladder, and hepatocystic triangle. These annotations were used as ground truth to train the AI model. The annotations aimed to delineate areas safe for dissection (Go zones) and areas where dissection could result in bile duct injuries (No-Go zones).
- (3) *Model training and validation.* The AI model underwent rigorous training using a tenfold cross-validation technique. This method ensured that the model's performance was robust and generalizable. The primary metrics for evaluating the model's performance included the Intersection-Over-Union (IOU) and F1 score, which measure the spatial overlap and accuracy of the predicted zones compared to expert annotations.

Performance metrics. The evaluation of GoNoGoNet demonstrated promising results:

- (1) For Go zones, the model achieved a mean accuracy of 0.94 and specificity of 0.94, indicating a high likelihood of correctly identifying safe areas for dissection. For No-Go zones, the accuracy was 0.95, with a sensitivity of 0.80, reflecting the model's effectiveness in identifying dangerous areas to avoid.
- (2) The mean IOU for Go zones was 0.53, and for No-Go zones, it was 0.71. The F1 scores were 0.70 for Go zones and 0.83 for No-Go zones, indicating a good spatial correlation between the AI predictions and expert annotations.
- (3) The model's inference time was less than 0.01 seconds per frame, satisfying real-time requirements and making it feasible for intraoperative guidance.

Impact on surgical practice. GoNoGoNet's ability to provide real-time visual cues has several implications for surgical practice:

- (1) *Enhanced decision-making.* By clearly marking safe and dangerous zones, GoNoGoNet aids surgeons in making informed decisions during dissection, potentially reducing the incidence of bile duct injuries.

- (2) *Training and education.* The algorithm serves as a valuable tool for surgical training, helping novice surgeons understand and visualize critical anatomical structures and safe dissection zones. This can enhance their learning curve and improve overall surgical proficiency.
- (3) *Patient safety.* The use of GoNoGoNet can lead to improved patient outcomes by minimizing surgical complications, thus enhancing the overall safety and efficacy of LC.

Conclusion. GoNoGoNet represents a significant advancement in the application of AI for intraoperative decision-making and surgical education. Its ability to accurately identify anatomical structures and delineate safe and dangerous zones during LC demonstrates its potential to augment surgical performance and improve patient safety. As AI technology continues to evolve, the integration of such tools in surgical practice is likely to become increasingly prevalent, offering new possibilities for enhancing surgical outcomes and education.

Objective evaluation of surgical performance using artificial intelligence

Traditional methods of assessing surgical skills often rely on expert opinion, which can be subjective, time-consuming, and costly. AI offers a solution by providing automated, objective, and consistent assessments. A systematic review by Pedrett and colleagues demonstrates the effectiveness of AI in evaluating technical skills in minimally invasive surgery.¹⁶⁵ The review analyzed 50 studies that employed AI models to assess surgical performance using motion data from surgical videos and kinematic data from robotic systems. These models, predominantly utilizing deep learning techniques, showed high accuracy in predicting skill levels in simulated environments.

The Johnson and Johnson Crowd-Sourced Assessment of Technical Skills (C-SATS)¹⁶⁶ is a video-based surgical assessment platform designed to improve surgical proficiency. It provides performance analytics, self-learning opportunities, and peer-to-peer interactions. Surgeons can upload their surgical videos to receive AI-powered insights, expert feedback, and crowd-sourced quantitative reviews. The platform supports continuous improvement by offering personalized feedback, tracking performance trends, and facilitating knowledge sharing among surgeons. C-SATS aims to enhance surgical skills, reduce operative risks, and ultimately improve patient outcomes.

Surgical Autonomy Program (SAP)¹⁶⁷ developed at the Duke University Department of Neurosurgery, uses a responsive web application to assess surgical competence, provide feedback, and visualize progress for surgical trainees in real-time. It is based on Vygotsky's Social Learning Theory and utilizes a TAGS scale for competency-based surgical education.

The Ethicon AI-powered Laparoscopic Skills Training Platform¹⁶⁸ integrates AR and AI to enhance laparoscopic surgical training. It offers real-time data, tactile feedback, and personalized learning experiences, allowing surgeons to hone essential skills through intelligent assessments and feedback. Surgical AI Trainer¹⁶⁹ evaluates surgeons' techniques using AI to analyze videos of real-life surgeries. It generates interactive reports that highlight skill gaps and provide targeted feedback to improve surgical outcomes. Developed by Dani Kiyasseh and his team, it focuses on making performance evaluations more accessible and effective. These platforms leverage advanced technologies like AI, AR, and video analysis to provide objective and actionable feedback, aiming to enhance surgical training and improve patient outcomes.

Postoperative performance analysis

AI-based systems for postoperative performance analysis typically involve the use of CV and ML algorithms to process and evaluate surgical videos. One prominent example is the Surgical AI System¹⁷⁰ (SAIS), developed by researchers at Caltech and Keck Medicine of the University of Southern California. SAIS uses AI to provide surgeons with real-time, objective feedback on their

performance by analyzing surgical videos. It utilizes a combination of RGB frames and optical flow data to capture both spatial and temporal aspects of surgical activity. The spatial features are extracted using a vision transformer pretrained on large datasets like ImageNet, while the temporal relationships between frames are captured through an attention mechanism that highlights the most relevant frames for skill assessment.

To enhance the reliability of these assessments, the SAIS incorporates a module called Training With Explanations (TWIX).^{171, 172} This module uses human-generated explanations as a supervisory signal to train the AI system, ensuring that the AI's focus aligns with expert evaluations. The training process involves annotating video samples with segments of time deemed important for skill assessments based on a predefined set of criteria. These annotations guide the AI system in learning to identify and emphasize critical moments in the surgical procedure that reflect the surgeon's skill level. By integrating human expertise into the training process, TWIX improves the accuracy and fairness of the AI-generated feedback.

The implementation of AI systems like SAIS for postoperative performance analysis has demonstrated significant improvements in the accuracy and reliability of surgical skill assessments. Studies have shown that AI-based explanations often align closely with human expert evaluations, providing consistent and actionable feedback to surgeons. For instance, in the assessment of needle handling and needle driving skills during surgical procedures, SAIS enhanced with TWIX achieved higher accuracy and reliability compared to assessments without the TWIX module. The precision-recall curves indicated that the AI-generated explanations were not only reliable but also reduced the variability in assessments across different surgeon subcohorts.

Moreover, the use of AI for performance analysis has proven effective in identifying and mitigating biases in feedback. In traditional assessments, biases can arise due to subjective judgments and inconsistencies in evaluations. However, AI systems trained with TWIX have shown a marked reduction in explanation bias, ensuring that feedback is equally reliable for novice and experienced surgeons. This equitable feedback is crucial for maintaining a fair and effective training environment, where all surgeons have the opportunity to improve their skills based on objective and consistent evaluations.

Several case studies highlight the practical applications and benefits of AI-driven postoperative performance analysis. For example, the implementation of SAIS in multiple hospitals, including the University of Southern California, St. Antonius Hospital in Germany, and Houston Methodist Hospital, demonstrated its effectiveness in diverse clinical settings.¹⁷² The system's ability to generalize across different hospitals and maintain high reliability in skill assessments underscores its potential for widespread adoption. These case studies also revealed that AI-based assessments could provide detailed feedback on specific aspects of surgical performance, such as the smoothness of needle driving or the precision of tissue handling, which are critical for surgical success.

In training environments, AI systems are being used to provide real-time feedback to surgical trainees. For instance, medical students practicing suturing techniques on gel models of the bladder and urethra received feedback from SAIS, which closely aligned with expert evaluations. This real-time, AI-generated feedback helped trainees identify areas for improvement and refine their techniques more effectively than traditional training methods. The ability to provide consistent and detailed feedback regardless of the trainee's location or the availability of expert mentors makes AI an invaluable tool in modern surgical education.

Robotic surgery and artificial intelligence-assisted guidance

Intraoperative enhancements

AI plays a pivotal role in enhancing intraoperative procedures, particularly through real-time image enhancement techniques. AI algorithms are employed to process and improve the quality of intraoperative images, facilitating better visualization of anatomical structures and surgical

instruments. For example, an online preprocessing framework for knee arthroscopy has been developed, incorporating denoising, deblurring, and color-correcting of real-time camera imaging, which significantly enhances the clarity of the surgical field.¹⁷³ Another innovative application utilizes CNN combined with transformers to remove surgical smoke, ensuring a clear view during procedures that involve electrocautery.¹⁷⁴

Native tissue recognition

AI is also enhancing the capabilities of surgical robots in tissue identification and differentiation. For instance, deep learning models developed by Kumazu and colleagues¹⁷⁵ can segment loose connective tissue fibers to establish safe dissection planes during robot-assisted gastrectomy, thereby reducing the risk of damaging critical structures. In surgical oncology, AI models utilizing fiber-based fluorescence lifetime imaging generate heat maps indicating probable cancer locations, aiding surgeons in performing more precise tumor excisions.¹⁷⁶ These advancements in tissue recognition are particularly vital for procedures like brain tumor removal, where accurately distinguishing between malignant and healthy tissue is crucial for successful outcomes.¹⁷⁷

Tactile feedback and instrument delineation

One of the challenges in robotic surgery is the absence of tactile feedback, a critical component in traditional open surgery. Recent advancements have introduced haptic feedback mechanisms that give surgeons a sense of touch. For instance, systems providing vibratory feedback to simulate tissue resistance have been shown to reduce the force exerted during surgical tasks, thereby preventing tissue damage.¹⁷⁸ Additionally, AI-enhanced instrument delineation plays a crucial role in accurately identifying and differentiating surgical instruments within the operative field. This capability is particularly important for augmented reality applications in surgery, where precise overlay of digital information on the surgical field can greatly assist the surgeon.¹⁷⁹

Stepwise automation

AI is significantly advancing intraoperative procedures by automating specific surgical tasks and reducing the physical and cognitive workload on surgeons. One notable development is the creation of algorithms for autonomous camera control in the da Vinci surgical system. These algorithms optimize camera positioning and zoom levels based on the surgical context, thereby improving visualization and efficiency during surgery.¹⁸⁰ Further advancements include the automation of suturing tasks, where AI models guide the robotic system through the complex steps of surgical suturing. This significantly enhances efficiency and precision, as seen in fully automated laparoscopic bowel anastomoses using CNNs coupled with U-Net algorithms, representing a significant leap towards higher levels of surgical automation.¹⁸¹

Skill assessment to improve education

In a study conducted by Fard and colleagues¹⁸² 8 global movement features were collected from surgeons with varying robotic experience while performing 2 minimally invasive surgical tasks using the da Vinci robot. Kinematic data from both hands during surgical knot tying and suturing was analyzed, and global rating scores were created to group the surgeons into 2 skill

levels: novice and expert. Three different ML techniques for classification were applied to automatically predict robotic surgical skill level. The quantitative metrics were found to be the most relevant for correct classification and the predictive model was able to correctly classify surgeons with 82.3% and 89.9% accuracy for knot tying and suturing, respectively.

In another study, Lavanchy and colleagues¹⁸³ developed a multi-stage ML algorithm to automate surgical skill assessment using video footage of LC procedures. Good surgical performance was determined by narrow and focused instrument handling within the operating field and poor surgical performance was indicated by frequent changes of instrument direction within a larger field. With an 87% accuracy, this novel method was able to identify and analyze the movement pattern of surgical tools to distinguish good and poor surgical performance.

Technology and global surgical education

Low fidelity simulations for low- and middle-income countries

Over the past decade, the development of AR/VR in surgery has predominantly focused on high-income settings, often neglecting more affordable applications, especially for VR. AR development has also required substantial network connectivity and financial resources for data storage and transfer. Additionally, VR advancements have primarily addressed the needs of high-income countries, where open procedures are less common. Consequently, VR has not been adapted for open procedures, which are still prevalent in cancer operations within resource-constrained settings. Major surgical and cancer surgery commissions have highlighted that the most significant gaps in global surgery disproportionately affect the poorest countries.

Researchers developed a low-fidelity VR surgical simulation model to enhance surgical oncology capacity in low-resource settings.¹⁸⁴ The team used commercially available technology, including standard VR software and Oculus Rift hardware, to create a cost-effective and immersive training environment. The VR system was designed to replicate an operating room in a 1:1 scale, incorporating detailed models of surgical instruments and internal anatomy. This setup allowed for realistic simulations of surgical procedures, specifically focusing on radical abdominal hysterectomy for cervical cancer. The visuals maintained a frame rate greater than 60 frames per second to ensure high visual fidelity and immersion. Training modules were co-designed by experts in various fields, including learning sciences, human behavior, VR, and gynecologic oncology, to ensure the simulation was both educationally effective and engaging.

The results demonstrated that this low-cost VR simulation could effectively prepare novice surgeons for complex surgical procedures. The simulation was tested for its ability to accelerate the acquisition of surgical skills, with participants showing significant improvement in their technical proficiency. By using this VR model, the researchers were able to reduce the time and cost associated with traditional surgical training methods. The simulation's potential for scalability and adaptability makes it a promising tool for increasing surgical oncology capacity in low- and middle-income countries, where access to advanced surgical training is often limited. If validated in real-world clinical trials, this approach could significantly impact global surgical education and training, particularly in resource-limited settings.

Julian Varas and his team developed a low-cost, Objective Structured Clinical Examination (OSCE) scenario for training surgical skills in trauma surgery.¹⁸⁵ The method involved creating 5 simulation-based stations: bowel anastomosis, vascular anastomosis, penetrating lung injury, penetrating cardiac injury, and gastric perforation (laparoscopic suturing). These stations used ex vivo porcine models and commercially available simulation tools. The training was conducted over a 3-day boot camp with 75 participants, including surgical residents and recently graduated surgeons from various programs. Each procedure was videotaped and assessed by experts using a modified Objective Structured Assessment of Technical Skills (OSATS) and a task-specific rating scale (SRS). Additionally, participants' self-confidence was evaluated before and after the training using a Likert-type survey.

The results showed significant improvements in participants' technical skills and self-confidence across all stations. Statistically significant increases in both OSATS and SRS scores were observed post-training, particularly in the cardiac and pulmonary stations where initial pretraining scores were the lowest. Interestingly, the time to complete procedures increased for pulmonary tractotomy, bowel anastomosis, and vascular anastomosis stations, indicating a learning curve effect where participants took more time to perform the procedures accurately. Conversely, time for bleeding control and ventricular repair in the cardiac injury station decreased significantly, reflecting improved efficiency. The study highlighted the effectiveness of this OSCE-based training program in enhancing trauma surgical skills, particularly in low-resource settings, and suggested the need for early incorporation of simulation-based tools in surgical residency programs to better prepare surgeons for high-stakes trauma scenarios.

In another study researchers developed cost-effective, low-fidelity surgical simulation models to enhance pediatric surgical training in low-resource settings.¹⁸⁶ The models created include neonatal bowel anastomosis, duodenoduodenostomy for discrepancy anastomosis, gastrostomy, and pyeloplasty. These models were constructed using animal tissue sourced from local abattoirs and basic surgical instruments. The simplicity and low cost (each model costing less than \$10) make these simulations accessible and reproducible in resource-limited healthcare environments.

The methods involved detailed steps for each surgical procedure. For example, the bowel anastomosis model used bisected segments of sheep intestine to simulate the procedure, while the duodenoduodenostomy model employed half a sheep bladder and a segment of sheep bowel to replicate the surgery. These models were utilized in 9 courses run by the International Affairs Committee of the British Association of Pediatric Surgeons across Africa since 2008. The success of these models in training sessions indicates their potential to provide sustainable training programs that enhance surgical skills and improve patient care in low- and middle-income countries.

The results of implementing these low-fidelity models showed significant promise in improving the practical skills of surgical trainees. The models provided a valuable training tool that could be easily integrated into existing programs, ensuring that trainees gain essential surgical skills despite limited resources. The study underscores the importance of accessible and cost-effective training solutions to address the global need for competent surgical care, particularly in settings where high-fidelity simulations are not feasible due to financial and logistical constraints.

Telemedicine

Telemedicine has revolutionized various aspects of health care, and its impact on surgical training and education has been particularly significant.¹⁸⁷ By leveraging technology to bridge geographical gaps, telemedicine provides surgical trainees with access to expert guidance, real-time feedback, and immersive learning experiences regardless of their location.

Remote mentoring and telementoring

Telemedicine facilitates remote mentoring, where experienced surgeons guide trainees through surgical procedures in real time using video conferencing and augmented reality.¹⁸⁸ This approach allows trainees to perform operations with expert oversight, reducing the risk of errors and enhancing learning outcomes. Telementoring has been particularly beneficial in resource-limited settings where access to specialized surgical expertise is limited. Previously described platforms like C-SATS¹⁶⁶ use telemedicine to allow peer reviews and expert evaluations of surgical videos, providing valuable feedback to trainees.

Continuing medical education (CME)

Telemedicine supports CME by providing access to online courses, webinars, and virtual conferences.¹⁸⁹ Surgeons can stay updated with the latest advancements in surgical techniques and technologies through these platforms, ensuring lifelong learning and professional development. The ACS offers telemedicine-based CME programs to enhance the knowledge and skills of practicing surgeons.¹⁹⁰

Several studies and initiatives have demonstrated the effectiveness of telemedicine in enhancing surgical training and education. A study evaluated the use of telementoring in laparoscopic surgery.¹⁹¹ The researchers used telemedicine to connect trainees with expert mentors who provided real-time guidance during surgeries. The study reported significant improvements in the trainees' performance and a reduction in procedural errors, demonstrating the efficacy of telementoring in surgical training. Additionally, the Global Initiative for Children's Surgery has implemented telemedicine-based training programs in low-resource settings.¹⁹² These programs use telemedicine to deliver lectures, conduct workshops, and provide hands-on training through remote mentoring. The initiative has successfully trained numerous pediatric surgeons in developing countries, improving surgical care for children worldwide.

Research consistently shows that telemedicine enhances surgical training by providing access to expert knowledge, reducing the learning curve for complex procedures, and improving surgical outcomes. Trainees benefit from real-time feedback, increased exposure to diverse cases, and the ability to practice in a risk-free environment. Telemedicine reduces the costs associated with traditional surgical training, such as travel and accommodation expenses for trainees and trainers. By delivering training remotely, telemedicine ensures that high-quality education is accessible to a broader audience, including those in underserved regions. Studies have shown that telemedicine-based training leads to better retention of surgical skills.^{189,191} The use of simulations and repetitive practice under expert supervision helps trainees internalize techniques and develop muscle memory, leading to long-term competency.

Importance of global multidisciplinary partnerships

The integration of AI into global surgical practices necessitates a comprehensive and collaborative approach. Multiple stakeholders, each with distinct perspectives and responsibilities, shape this process.¹⁹³ Policymakers provide essential governance, ensuring AI interventions align with global health care goals. Health care professionals—surgeons, nurses, technicians—are at the forefront, integrating AI tools and advocating for patient needs. Local community organizations, including faith-based groups, can provide vital cultural perspectives and build trust during implementation. Central to these efforts are the patients themselves, whose experiences and feedback ensure that AI remains patient centric. Tech developers and researchers pioneer AI advancements, while industry partners bridge the gap between innovative solutions and real-world applications. By providing necessary resources, fostering collaborations, and ensuring market access, they ensure that AI tools reach the regions and populations that need them the most. These stakeholders form the backbone of the AI surgical ecosystem, and incentivizing their alignment is paramount.

An example of exceptional collaboration is evident at Rutgers University, where a unique initiative called the Simulation Collaborative Advancing Research Learning Education and Technology (SCARLET) has been launched. This model aims to consolidate simulation efforts across the university's expansive footprint, including the New Jersey Medical School (Figs. 21, 22), Robert Wood Johnson Medical School, and the Rutgers Dental and Nursing schools. The collaborative's specific goals are to advance multidisciplinary health professions simulation, minimize duplicative efforts, standardize educational initiatives, and foster research development.



Fig. 21. An attending surgeon teaching a medical student in the Year 1 Module 7 - Venipuncture and Peripheral IV from the American College of Surgeons/Association for Surgical Education (ACS/ASE) Medical Student Simulation-Based Surgical Skills Curriculum.



Fig. 22. Students learning minimally invasive surgery skills at the Rutgers New Jersey Medical School Basic Skills Lab.

Conclusions

There has been a tipping point over the last 2 decades regarding simulation. It is no longer acceptable to practice on patients or sacrifice animals to learn our craft when a model, simulator, or VR is an available option. Since the first publication in *Current Problems in Surgery*, simulation has moved from being a luxury of a relatively few medical schools to a standard for teaching and learning. Surgical educators have adopted simulation to teach basic skills and nontechnical skills in a safe learning environment. This monograph has reviewed the use of models, video trainers, simulation, and virtuVR simulators (Fig 20). We have introduced the role of AI to enhance the realism of simulation, facilitate training, guide decision-making, and improve patient safety.

As medicine advances and procedures become increasingly complex and novel, opportunities for repetitious training as a resident are limited. AI offers a technology-focused solution that gives surgeons an opportunity to test their abilities in performing new techniques while receiving real-time, objective feedback, in a controlled and accessible environment. However, the incorporation of AI into the existing training modalities is still a new endeavor and we currently have limited published data and limited curricula. Increasing the use of AI will provide more exposure for surgical trainees to practice and receive guidance on how to improve their procedural skills which will ultimately prepare them better for independent surgical practice.

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